Machine Learning Surrogates for Agent-Based Models in Transportation Policy Analysis

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Abstract

Effective traffic policies are crucial for managing congestion and reducing emissions. Agent-based transportation models (ABMs) offer a detailed analysis of how these policies affect travel behaviour at a granular level. However, computational constraints limit the number of scenarios that can be tested with ABMs and therefore their ability to find optimal policy settings.

In this proof-of-concept study, we propose a machine learning (ML)-based surrogate model to efficiently explore this vast solution space. Leveraging a Graph Neural Network (GNN), the model predicts the effects of traffic policies on the road network at the link level.

We implement our approach in a large-scale MATSim simulation of Paris, France, covering over 30,000 road segments and 10,000 simulations, applying a policy involving capacity reduction on main roads. The ML surrogate model achieves an overall R^2 of 0.76, with significantly lower Mean Squared Error and Mean Absolute Error than naive baselines. On primary roads where the policy applies, it reaches an R^2 of 0.95. This study therefore shows that GNNs can act as surrogates for complex agent-based transportation models with the potential to enable large-scale policy optimization, helping urban planners explore a broader range of interventions more efficiently.

Keywords: Graph Neural Networks, Agent-based simulations, Optimization-Based Transportation Policies, Surrogate Models, Street Capacity Reduction

1. Introduction

Cities around the world face growing challenges in terms of congestion and air pollution, exacerbated by rapid urbanization and population growth. Addressing these issues requires effective policies that reduce car dependency while maintaining flexibility for residents.

Urban planners have a wide range of policy interventions at their disposal, including congestion charging, parking and traffic control measures, the establishment of limited traffic zones or rededicating space allocated for cars. In recent years, additional mobility options such as shared mobility and micromobility have emerged, expanding the set of possible policy measures.

However, designing effective policies is highly complex. Urban planners must not only decide which policies to introduce but also determine the optimal level at which they should be implemented and, for most measures, their optimal spatial extent. The number of possible policy combinations grows exponentially, leading to an immense solution space. This multi-dimensional space is too vast to be explored manually based on experience alone. In practice, stakeholders often limit the solution space to a few scenarios, which in all likelihood will not include the optimal solution, and in the worst case might not even be close to it. Therefore, a more structured approach is needed to narrow down the most promising solutions.

Agent-based simulations are commonly employed to assess how potential policy measures impact the broader transportation system. While these models provide detailed and realistic assessments of the effect of policies in terms of congestion, emissions, and travel times per mode, their computational

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intensity severely limits the number of scenarios that can be tested in practice.

Agent-based models can be understood as functions that map policy interventions to their corresponding effects - both at the link level, such as changes in traffic volumes on specific street segments, and at the system level, including emissions, mode shifts, and travel times.

Machine learning (ML) has emerged as a powerful tool for approximating complex functions when trained on sufficient data. A key milestone in its application to transportation was the study by Sroczyński and Czyżewski (2023), which demonstrated that ML models can replicate traffic patterns with accuracy comparable to microscopic simulations. While not directly related to this work, such advances highlight the potential of ML to approximate intricate relationships in transportation systems.

If machine learning could effectively approximate agent-based models, it could serve as the foundation for fast and scalable surrogates. Even if their predictions were not perfectly aligned with full agent-based simulations, the ability to rapidly filter out ineffective policy setups and thus being able to focus computational resources on the most promising solutions would be a major advantage. Furthermore, such an approach could reveal novel policy combinations that human planners might not have initially considered. Once trained, such a surrogate model could predict the impacts of policies within seconds, enabling efficient scenario exploration and large-scale optimization.

Graph Neural Networks (GNNs) are machine learning models specifically designed to handle graph-structured data. GNNs have gained increasing attention in the transportation domain due to their ability to capture spatial dependencies, making them well-suited for modeling complex network dynamics. As such, intersections or street segments can be represented as nodes, while edges capture relationships such as connectivity or proximity. Their ability to capture spatial dependencies within urban networks makes them an intuitive choice for predicting the impact of policy interventions on link-level.

In this paper, we present a surrogate model for agent-based transport simulations, leveraging a Graph Neural Network to approximate the effects of policy interventions. We evaluate its performance using the large-scale MATSim simulation of Paris, France, applying the policy "50% capacity reduction on main roads". Importantly, the developed method is not limited to this specific policy or location; it can be applied to any policy intervention in any city.

Our results show that the surrogate model can accurately predict changes in traffic volume per street segment resulting from the policy, demonstrating the feasibility of using Graph Neural Networks as surrogates for agent-based models in transport planning. This approach enables efficient solution-space exploration, allowing (1) planners to systematically narrow down policy options and identify the most effective interventions and (2) providing an efficient methodology for objective evaluation in optimization-based approaches.

This paper is structured as follows: Section 2 reviews the relevant literature, followed by Section 3, which introduces the used methodology. Section 4 provides an in-depth explanation of the surrogate model, while Section 5 outlines the methodology for assessing its accuracy. In Section 6, we introduce the case study based on a MATSim simulation of Paris, France, to validate the proposed approach. Finally, Section 7 presents the results, and Section 8 concludes with key insights, methodological limitations, and directions for future research.

2. Background

2.1. Agent-Based Models for Transport Policy Evaluation

Agent-based models (ABMs) have become a widely adopted approach for analyzing transportation systems due to their ability to simulate individual decision-making processes and interactions at a fine-grained level (Kagho et al., 2020). Unlike macroscopic and mesoscopic models, which aggregate traveler behavior, ABMs explicitly model each agent (e.g., travelers, vehicles) and their adaptive choices, allowing for a more detailed representation of network dynamics and behavioral responses to policies, as shown e.g. by Horni et al. (2016) and Macal and North (2010).

Agent-based models have been widely applied to assess various transportation policies, including congestion pricing (Ben-Dor et al., 2024), parking fee schemes (Balac et al., 2017), the impact of speed limits on street safety and emissions (Lu et al., 2023), and to evaluate low-traffic zones (Yin et al., 2024; Müller et al., 2023).

ABMs are particularly well-suited for analyzing the impact of emerging mobility solutions and their integration into existing transportation systems. As urban mobility continues to evolve, there is an urgent need for innovative transport solutions that not only improve efficiency and sustainability but also integrate seamlessly with current infrastructure.

The integration of emerging transport modes and mobility schemes into existing transportation systems presents both challenges and opportunities for policymakers and planners. Understanding travel behavior and demand in these evolving systems is essential for designing adaptive infrastructure and policies that can accommodate technological advancements and urban mobility transitions. Modern travelers' decisions are influenced by spatial, societal, and economic factors, adding complexity to the prediction of travel behavior in changing environments. Recent studies emphasize the need to analyze how emerging transport modes - such as ridesharing, electric vehicles, and autonomous vehicles - interact with existing systems. These interactions can be competitive or complementary, depending on regional and national contexts (Bastarianto et al., 2023).

Agent-based models provide a powerful approach to studying these interactions. They can be used to assess the integration of new transport modes with existing systems and to evaluate sustainable mobility transitions, such as shared mobility services (Hörl et al., 2021), electric vehicle adoption (Marmaras et al., 2017), and demand-responsive transit (Auld et al., 2016).

By simulating traveler behavior and mobility trends, ABMs enable researchers and policymakers to explore potential impacts and identify strategies for optimizing urban mobility. Despite these advantages, the increasing complexity of ABMs also poses computational challenges, especially when scaling to large urban regions. As the number of agents and interactions increases, computational costs rise rapidly, making real-time policy testing difficult (Balmer et al., 2008).

2.2. Computational Challenges of Agent-Based Models

The disadvantage of agent-based simulations is a high run-time of usually several hours, limiting the amount of scenarios that can be tested. Large-scale ABMs simulate thousands to millions of agents, each with their own decision-making processes. This requires significant memory and processing power, with computational demands growing rapidly as more agents and decision rules are incorporated. Since many ABMs do not parallelize efficiently, execution times often increase non-linearly, making real-time scenario testing infeasible (Auld et al., 2016; Bastarianto et al., 2023).

Improving the scalabilitiy and efficiency of agent-based traffic simulations is an open research question. For the MATSim simulator, Laudan et al. (2025) compare local parallelization with distributed computing approaches, but show limitations in acceleration. Furthermore, MATSim is a highly modular framework that is used by numerous domain experts from the transport field that often only have basic programming knowledge. This is a main reason why MATSim is still implemented in Java which is comparably slow with respect to other programming languages, but allows for quick adaptation and extension with new components. An interesting approach, therefore, remains to provide the framework with a high degree of implementation flexibility and find ways to approximate the high-level model outcomes. Smilovitskiy et al. (2025) present a GPU-accelerated transport model for the Isle of Wight using the FLAME-GPU framework. By comparing it with traditional non-GPU simulations like SUMO, the study demonstrates significant performance gains, enabling the simulation of larger vehicle populations and more detailed traffic interactions. However, the computational gains do not fully eliminate scalability issues, particularly for real-time or city-scale applications. Despite leveraging GPU hardware, simulations of large-scale networks still require extensive resources, and the dependency on specialized hardware limits accessibility. Furthermore, GPU-accelerated simulations often focus on the specific task of traffic assignment, but do not cover other aspects such as multimodality or mode choice behavior. An example where such elements are taken into account is presented by Saprykin et al. (2019). However, the rigid programming requirements for the GPU interaction limit modularization and easy adaptation and extension of the software by domain experts with basic programming knowledge as explained above. Manley et al. (2014) introduce a hybrid agent-based modeling framework that balances behavioral realism with computational efficiency by integrating detailed driver behavior with a simplified collective traffic flow model. They demonstrate how this approach enables scalable simulations of urban traffic dynamics while improving computational performance compared to traditional agent-based methods. However, they still face trade-offs between behavioral realism and scalability. The challenge lies in ensuring that large-scale simulations maintain sufficient fidelity in representing individual decision-making and emergent traffic phenomena.

One widely-used alternative to speed up agent-based models for practical applications, is downsampling the population size. For example, Llorca and Moeckel (2019) analyzed the effects of population scaling in agent-based transport models, investigating how different scale factors impact model runtime, travel times, and link volumes. The authors conclude that scaling down agent populations reduces runtimes but must be done carefully to avoid distortions in travel times. For the Munich metropolitan area they found that a 5% sample produces travel time distributions similar to a full-scale simulation (100% sample) while being 50 times faster. Further, they find that the choice of scale factor depends on the analysis level: aggregate studies may tolerate smaller samples, while detailed corridor or small-area analyses may require full populations.

However, downsampling the population size in agent-based transport models increases the inherent stochasticity due to the random selection of agents, which lead to variability in simulation outcomes. This variability arises because different subsamples may not accurately represent the full population's travel behaviors, affecting demand distribution, route choices, congestion levels, and travel times. Such stochastic effects are particularly pronounced with smaller sampling rates, where statistics can substantially deviate from those of full-scale models. (Ben-Dor et al., 2021)

Concluding, the goal of fully scalable agent-based models remains out of reach, as current research has yet to make them feasible for large-scale applications. In practice, population downsampling is used to reduce computational demands, but even with this approach, testing a single policy can take days. Consequently, the demand for more computationally efficient alternatives of agent-based models has driven interest in machine learning-based surrogates, which could approximate agent-based model outcomes while preserving key behavioral dynamics, enabling faster policy evaluation and optimization.

2.3. Machine Learning Surrogates for Traffic Simulations

Machine learning surrogates for traffic simulations learn the relationships between a simulations' input parameters (characteristics of the different street segments, due to its nature or due to the introduced policy) and output variables (traffic flow metrics on street segment level) from data generated by the traffic simulation. Once trained, the surrogate model can rapidly predict the outcomes of various policy scenarios without the need for computationally intensive simulations.

Table 1 provides an overview for machine learning surrogates for traffic simulations: Liu et al. (2020) propose a machine learning and particle swarm optimization (PSO) approach to efficiently calibrate microscopic traffic simulation parameters. Four machine learning models (Decision Tree, SVM, Gaussian Process Regression, and ANN) are tested to predict simulation outputs, with ANNs achieving the best accuracy. The ANN-based surrogate model is then embedded in PSO to optimize parameters, demonstrating superior efficiency and effectiveness when applied to TransModeler with field data from Shanghai. While effective, the authors recognize some limitations: They used a Multilayer Perceptron (MLP) for the artificial neural network but suggest that future studies explore Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs) for better performance.

Cervellera et al. (2021) investigate the use of a deep learning architecture with low-discrepancy sampling for surrogate modeling of traffic simulations. The approach leverages deep neural networks for their strong approximation capabilities and low-discrepancy sequences to efficiently cover the input space. Applied to the SUMO microsimulator, this method outperforms traditional surrogate models, such as Gaussian processes, in high-dimensional optimization tasks like traffic demand calibration and traffic light optimization. The authors highlight the need for a deeper theoretical analysis of their approach in the surrogate modeling framework and suggest broader testing across different applications beyond traffic simulation.

Graph Neural Networks have gained increasing attention in the transportation domain due to their ability to capture spatial dependencies, making them well-suited for modeling complex network dynamics. As a class of deep learning models designed for graph-structured data, GNNs represent entities such as intersections or street segments as nodes, while edges encode relationships like connectivity or proximity. Their architecture enables the effective modeling of both local and global spatial dependencies, which has led to successful applications in various traffic-related tasks. For example, Jiang and Luo (2024) provide a comprehensive review of GNNs in traffic forecasting.

Despite their potential, the use of GNNs as surrogates for traffic simulations remains largely unexplored. To the best of the author's knowledge, only Narayanan et al. (2024) have investigated this application: In their study, they examine the feasibility of GNNs as surrogate models for the four-step model. They introduce a formal definition of a GNN-based surrogate and propose an augmented data generation process using small networks (15–80 nodes) for training. The authors evaluate different GNN architectures at the link level through classification and regression tasks. The classification task involves predicting traffic flow categories: 0–10 veh/h (no transport usage), 10–500 veh/h (low transport usage), and more than 500 veh/h (medium/high transport usage). The regression task focuses on estimating exact traffic flow values. While the results are promising, the study is constrained by its small scale and exclusive focus on the four-step model.

Authors	Model	Use Case	ML Method	Data Sources	Evaluation Metrics	Key findings
Liu, Zou, Ni, Gao, Zhang (Liu et al., 2020)	Trans - Modeler (microscopic model)	Traffic simulation calibration	Decision trees, Support Vector Machines, Gaussian Process Regression, Artificial Neural Networks (ANN), Particle Swarm Optimization (PSO)	Field data, simulated outputs	Prediction accuracy, computational efficiency	ANNs yielded the best prediction accuracy; ML+PSO methodology improved computational efficiency and effectiveness.
Cervellera, Maccio, Rebora (Cervellera et al., 2021)	SUMO	Origin-destination (OD) demand calibration, traffic light optimization, and strategic mobility planning.	Deep Neural Networks (DNNs), trained using low-discrepancy sampling (Sobol' sequences)	Data generated using SUMO, with OD flow values as inputs and simulated traffic flows as outputs. Medium-sized urban network.	1) Estimation accuracy of traffic flow at measurement points (MAE), 2) Distribution preservation comparing predicted and true flow distributions (MMD), 3) Surrogate model effectiveness in OD demand calibration.	1) Deep learning-based surrogates outperform Gaussian Processes in high-dimensional traffic simulations, 2) Low-discrepancy sampling (Sobol' sequences) improves training efficiency, accuracy, and reduces variance, 3) The surrogate better preserves traffic flow distributions, making it suitable for demand calibration, 4) Combining deep learning with low-discrepancy sampling enhances efficiency without sacrificing accuracy.
Narayanan, Makarov, Antoniou (Narayanan et al., 2024)	4-step model	Replicating traffic flows from simulations	GNN	Synthetic transport simulation data, networks consisting of 15-80 nodes	Accuracy (F1 Score, MAE and R^2) and computational efficiency	Proof-of-concept that a GNN can replicate traffic flows learned from the 4-step model.
This study	Generic ABM	Predicting the effects of transportation policies by approximating the outcomes of agent-based simulations	GNN	Simulated ABM outputs from a large-scale network comprising over 30,000 nodes	Accuracy (MSE, MAE, R^2, Pearson and Spearman Correlation), Impact of Simulation Stochasticity on the Error	Proof-of-concept that GNNs can learn the effect of transport policies from agent-based models

Table 1 – *Summary of Machine Learning Surrogates for Transport Simulations, sorted by year*

In our study, the surrogate model effectively captures the complexity of agent-based simulations. Additionally, we evaluate our approach on large-scale networks - specifically Paris, France, with 30,000 street segments. Finally, given that agent-based simulations inherently involve stochasticity due to the probabilistic nature of agent interactions, we examine its impact on model error - an aspect that has not been explored in previous studies.

2.4. Contribution of this study

In this work, we demonstrate that machine learning-based surrogate models can accurately replicate the outcomes of complex agent-based traffic simulations, effectively capturing the relationship between transportation policies and their impact on traffic volume at the link level.

Our findings highlight the potential of GNN-based surrogates for systematically exploring the vast solution space of traffic policy combinations and their extents, and it lays the foundation for large-scale policy optimization. The contribution of this study is three-fold:

1. Firstly, and at the core of this study, is the *Proof of Concept of a machine-learning based surrogate model for agent-based transport simulations.* By using a graph neural network architecture, we are able to predict the effects of policy interventions on a link level with high accuracy.

2. Secondly, we *examine the impact of simulation stochasticity on surrogate model error.* Agent-based models inhibit randomness, due to the probabilistic nature of agent-based interactions. In transportation, downsampling the population is a common strategy to reduce computational costs, but

increases stochastic variability. When the training data is generated from a downsampled population, the added randomness effectively acts as noise, potentially reducing the accuracy of the surrogate model. To better interpret the results, we analyze how the stochastic variability affects surrogate accuracy. This provides insights into the model's strengths and limitations, informing its applicability across different levels of downsampling.

3. Finally, we validate our methodology on a *large-scale case study.* The GNN surrogate is tested using MATSim with 1% downsampled data on the Paris network, encompassing over 30,000 streets. This demonstrates the effectiveness of the machine learning surrogate in capturing transportation dynamics.

3. Methodological Approach

Figure 1 gives an overview of the general methodology: The framework integrates an ABM with a GNN to predict changes in traffic volume at the link level under different policy scenarios. The process begins with an ABM that simulates travel behavior based on three key components: the street network, which represents the street infrastructure; travel modes, which define different transportation options; and a synthetic population, representing individual travelers and their mobility choices.

Two types of scenarios are considered: a *base case* (defined in Section 3.1), which captures existing traffic conditions such as baseline traffic volume $\overline{v}_e$ and positional attributes without any intervention, and *policy scenarios*, where the introduced policy leads to adapted traffic volumes.

To efficiently process this information, the street network is transformed into a *dual graph*, where street segments are represented as nodes, and their connectivity is defined by edges. A GNN-based surrogate model is then applied to capture relationships between the network and the applied policy, as described in Section 4. Finally, the trained GNN predicts the *change in traffic volume* at the link level under active policy conditions.

The relevant variables for the analysis are listed in table 2.

Symbol	Description
E	Set of edges in the street network, single edges denoted $e \in E$
R	Set of random seeds for which we perform simulation runs, random seeds denoted $\{r_1, r_2, ...\}$
P	Set of policies that may be implemented on edge level, policies denoted $\{p_1, p_2, ...\}$
v_e^r	Traffic volume on edge e in simulation run r, without policy intervention
$\overline{v}_e$	Average traffic volume on edge e, without policy intervention
$y_{e,p}^r$	Simulated *change* in traffic volume on edge $e \in E$ in simulation run r due to policy p.
$y_{e,p}$	Single realization of $y_{e,p}$: Simulated change in traffic volume on edge $e \in E$ due to policy p for one simulation run.
$\overline{y}_{e,p}$	Average simulated change of traffic volumes on edge $e \in E$ due to policy p: $\overline{y}_{e,p} = \frac{1}{R}\sum_R y_{e,p}^r$
$\overline{y}_p$	Mean of the simulated change of traffic volumes over all edges $e \in E$ due to policy p: $\overline{y}_p = \frac{1}{E}\sum_E \overline{y}_{e,p} = \frac{1}{E}\frac{1}{R}\sum_E\sum_R y_{e,p}^r$
$\hat{y}_{e,p}$	Predicted (GNN surrogate) change in traffic volume on edge $e \in E$ due to policy p.

Table 2 – *Symbols used in this analysis*

3.1. Base case

The *base case* represents a simulation scenario of the agent-based model without any policy intervention, serving as the reference point for predicting changes in traffic volume. It is generated by running multiple simulations with different random seeds and averaging the results at street level.

Consider the simulation, without any policy intervention, repeated with $|R|$ different random seeds. Each simulation repetition produces a traffic volume v_e^r, where $r = 1, 2, \ldots, |R|$. For a single edge $e \in E$, the base traffic volume for edge e is then defined by $\overline{v}_e$:

$$\overline{v}_e = \frac{1}{|R|}\sum_{r=1}^{|R|} v_e^r, \tag{1}$$

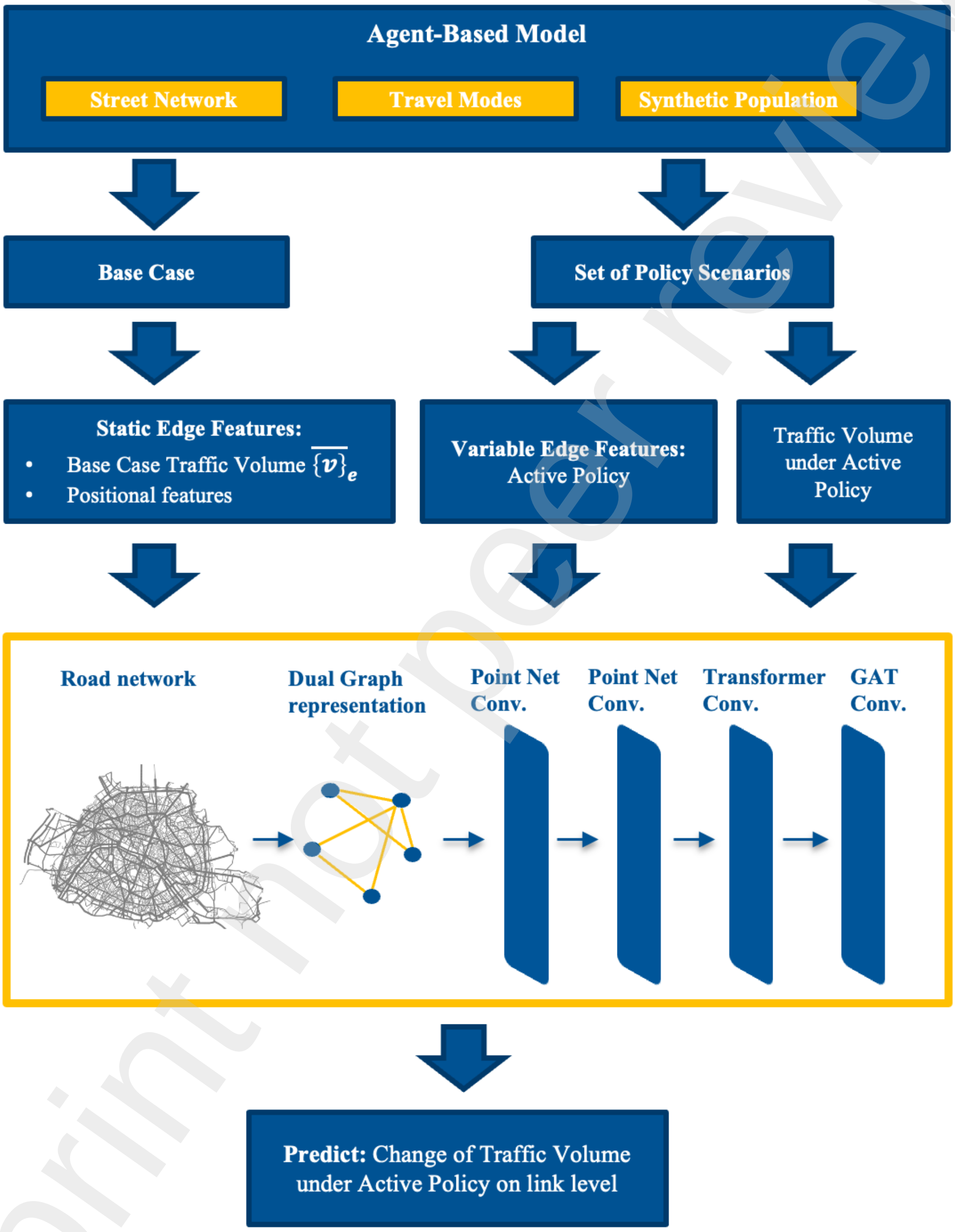


Figure 1 – *Overview over the Methodology*

With this, we can directly define the variance of the base case traffic volume for edge $e \in E$ as:

$$\sigma^2_{e,b} = \frac{1}{|R|} \sum_{r=1}^{|R|} (v^r_e - \overline{v}_e), \tag{2}$$

which quantifies the dispersion of traffic volume across independent simulation runs with different random seeds. In Section 5.3 we state an assumption under which this variance can be used for quantifying the aleatoric variance of our surrogate model.

3.2. Problem Statement

We aim to develop a machine learning-based surrogate model that estimates traffic volumes in response to policy interventions, using data from agent-based simulations. Specifically, the model learns how each intervention affects traffic on individual roads within the network.

Mathematically speaking, our objective is to learn a function F that maps a set of input simulation parameters P_{sim} to the corresponding simulation output S:

$$F(P_{sim}) \to S. \tag{3}$$

This defines a regression problem where the goal is to approximate F such that it captures the relationship between the input parameters and the resulting traffic volume changes. Let F_p denote F under policy intervention p, mapping input features to traffic volume changes under p. The goal is to learn the function:

$$F_p : \mathbb{R}^{|E| \times n} \to \mathbb{R}^{|E|}, \tag{4}$$

which maps a set of n street characteristics per edge to the predicted change in traffic volume $\hat{y}_{e,p}$ caused by a given policy intervention p. The impact of a policy intervention on a street is measured by the change in its traffic volume relative to the base case: For each edge $e \in E$, we predict the traffic volume change under intervention p compared to the base case.

To train the model, we use the Mean Squared Error (MSE) loss function, which quantifies the difference between predicted and actual values. Let x_e denote street characteristics for street e:

$$\mathcal{L}(F_p) = \frac{1}{|E|} \sum_{e \in E} (F(x_e) - \overline{y}_{e,p})^2 \tag{5}$$

$$= \frac{1}{|E|} \sum_{e \in E} \left(\hat{y}_{e,p} - \frac{1}{|R|} \sum_{r \in R} y^r_{e,p} \right)^2. \tag{6}$$

In an ideal setting, minimizing this loss would require $|R|$ simulation runs per policy intervention to compute the expected change in traffic volume. This distinction is crucial: the observed traffic volume change in a single run, $y^r_{e,p}$, differs from the expected change averaged over multiple runs with different random seeds, $\overline{y}_{e,p}$. Ideally, the surrogate model would predict $\overline{y}_{e,p}$ to account for randomness in the simulation.

However, due to computational constraints, each policy intervention is simulated only once ($|R| = 1$), so the predicted value $\hat{y}_{e,p}$ approximates $y_{e,p}$ rather than the expected mean. Consequently, we train our model using this single realization, minimizing:

$$\mathcal{L}(F) = \frac{1}{|E|} \sum_{e \in E} \left(\hat{y}_{e,p} - y^r_{e,p} \right)^2. \tag{7}$$

where $y^r_{e,p}$ is the observed traffic volume change for a simulation run with random seed r. As a result, the surrogate model approximates a single realization rather than the expected value, inherently incorporating the stochasticity present in the data. Since we always work with a single realization, we denote $y^r_{e,p}$ as $y_{e,p}$ hereafter.

The objective is to find the optimal function F^* that minimizes this loss:

$$F^* = \arg\min_F \mathcal{L}(F). \tag{8}$$

To develop an effective surrogate model, we first define the input and output parameters of the machine learning-based surrogate (see Section 4). The evaluation metrics are described in Section 5.1 and 5.2. Given the high computational cost of full-scale simulations, we train the model on downsampled population data, which increases the variance of traffic volumes across simulation runs with different random seeds. Since the model predicts a single realization of the stochastic simulation rather than the expected value $\overline{y}_{e,p}$, it inherently reflects the randomness in the data and cannot reduce stochasticity beyond what is already present. We further analyze this in Section 5.3, highlighting the role of base case variance in error estimation.

4. The Graph Neural Network Surrogate Model

As a theoretical basis for our surrogate model, we propose using Graph Neural Networks. These are machine learning models that deal with graph-structured data. Street networks of cities form a natural graph, where edges represent streets and nodes represent intersections. In our case, where the goal is to predict features on street level, GNNs are the intuitive choice. Because most GNNs are optimized for node-based predictions, we transform the street network graphs into dual graphs, where nodes correspond to street segments and edges capture the connections between segments. This transformation allows the GNN to operate at the level of street segments rather than intersections, making it well-suited for learning spatial dependencies and traffic dynamics. To provide a clear understanding, we first explain the model's inputs and outputs, followed by a detailed description of its architecture.

4.1. Input and Output

The input P_{sim} for the model is the dual graph containing the following attributes, for each street $e \in E$, where $|E|$ corresponds to the number of nodes in each of the transformed dual graphs:

1. *Static features* provide fixed attributes such as base traffic volume $\overline{v}_e$, capacity and speed limit in the base case, and street segment length.
2. *Variable features* represent attributes modified by implemented policies, such as reductions in capacity or speed.
3. *Positional features* capture the spatial information of each street, including the coordinates of its start and endpoint.

We denote the number of static, variable, and positional features, respectively, as d_s, d_v, and d_p.

The output S captures the impact of the implemented policy at the street level, such as changes in traffic volume relative to the base case, represented by $\hat{y}_{e,p}$ for each street e.

4.2. Architecture

The architecture of the GNN, shown in Figure 2, consists of multiple stages and supports a flexible number of features.

- *PointNet Convolution:* The model begins with two PointNet Convolution (Charles et al., 2017) layers. The first PointNet layer processes an input feature set of dimension $(d_p + d_s + d_v)$, mapping it to a higher-dimensional space using a local Multi-Layer Perceptron (MLP) with 256 hidden units. A global MLP follows, further transforming the feature representations with 512 hidden units in successive layers. The second PointNet layer expands the input dimensions to $512 + d_p$, integrating the coordinates for the end position of the street segment, and follows a similar MLP-based transformation.

- *Transformer Convolution:* Two layers of Transformer Convolutions (Shi et al., 2021) follow, leveraging self-attention mechanisms to dynamically assign importance to neighboring segments. The first Transformer layer operates with 64-dimensional embeddings (using four attention heads), while the second layer increases the embedding size to 128, again using four attention heads. This enables the model to capture long-range dependencies within the street network.

- *GAT Convolution:* The final stage consists of a two-layer Graph Attention Network (GAT) module (Veličković et al., 2017). The first GAT layer projects embeddings into a 64-dimensional space using attention-weighted aggregation. The final layer then reduces the feature space to a single output dimension, completing the model's predictions.

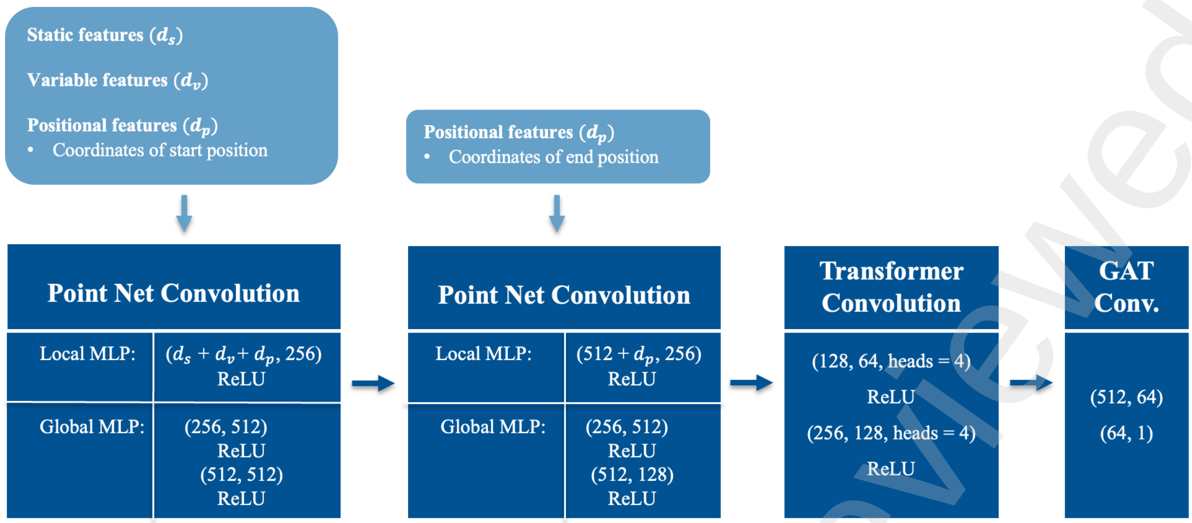


Figure 2 – *Architecture of the Graph Neural Network*

All layers (except for the ones in GAT module) use ReLU activations to introduce non-linearity. Kaiming and Xavier weight initialization methods are applied to ensure stable weight scaling, reducing the risk of vanishing or exploding gradients and improving training stability. The combination of PointNet, Transformer, and GAT convolutions enables the model to effectively capture both local segment interactions and global dependencies in the transport system, making it well-suited for learning from street network data. [1]

5. Methodology for Evaluating Surrogate Model Accuracy

While the model is trained by optimizing the MSE between simulated and predicted traffic volume changes, additional metrics are used to provide a more comprehensive assessment of its accuracy and robustness.

This section outlines our approach to evaluating the performance of the surrogate model on the link level. First, we introduce the evaluation metrics beyond MSE that are used for a thorough performance analysis. We also explain that model performance is assessed on the entire test set, with accuracy analyzed separately for different street types to identify variations (Section 5.1). Next, we describe the benchmarking methodology used to compare model performance against a naive baseline (Section 5.2). Finally, since agent-based simulations inherently introduce stochasticity, understanding its impact is crucial for a meaningful evaluation of the results. To better interpret the results, we examine the extent to which model error is influenced by simulation stochasticity (Section 5.3).

5.1. Evaluation Metrics

The test set consists of multiple scenarios, where each scenario corresponds to applying a policy p to a specific combination of districts. To evaluate model performance across different street types, we define aggregated evaluation metrics computed over the entire test set for each street type. Let S be the set of scenarios in the test set, and let T be the set of street types. Define E_t as the set of edges in the test set that belong to street type t. The total number of such edges is given by $|E_t| = |S| \times |T|$.

Aggregated Base Case Quantities

The *mean base case traffic volume* for street type t is given by:

$$\bar{v}_t = \frac{1}{|E_t|} \sum_{e \in E_t} \bar{v}_e, \tag{9}$$

where $\bar{v}_e$ represents the base case traffic volume for edge e (Section 3.1).

[1]The code for the GNN surrogate can be found on https://github.com/enatterer/gnn_predicting_effects_of_traffic_policies

The *base case variance of traffic volumes*, or simply *base case variance*, for street type t is defined as:

$$\sigma_{t,b}^2 = \frac{1}{|E_t|} \sum_{e \in E_t} \sigma_{e,b}^2, \tag{10}$$

where $\sigma_{e,b}^2$ is the variance of traffic volumes for edge e, as defined in Equation 2.

Aggregated Evaluation metrics

To evaluate model performance at the road-type level, we define the following quantities: The *average change in simulated traffic volume* for street type t under policy p is:

$$\bar{y}_{t,p} = \frac{1}{|E_t|} \sum_{e \in E_t} y_{e,p}, \tag{11}$$

where $y_{e,p}$ is the simulated traffic volume change for edge e under policy p. The *variance* for street type t under policy p is:

$$\sigma_t^2 = \sum_{e \in E_t} (y_{e,p} - \bar{y}_{t,p})^2. \tag{12}$$

We use the following metrics for assessing model performance:

1. *Mean Squared Error (MSE)*
   The MSE serves as the primary loss function during training. Although it is a standard regression metric, its squared term disproportionately penalizes large errors, making it more sensitive to outliers. The MSE for street type t under policy p is defined as:

$$MSE_{t,p} = \frac{1}{|E_t|} \sum_{e \in E_t} (\hat{y}_{e,p} - y_{e,p})^2. \tag{13}$$

2. *Mean Absolute Error (MAE)*
   To complement MSE, we report the Mean Absolute Error, which is more interpretable in practical applications as it represents the average absolute error in the same units as the input (veh/day). Additionally, MAE is more robust to outliers than MSE, as it treats all errors equally rather than amplifying large deviations. This makes it particularly useful in cases where occasional extreme values might otherwise dominate the error metric. The MAE for street type t under policy p is computed as:

$$MAE_{t,p} = \frac{1}{|E_t|} \sum_{e \in E_t} |\hat{y}_{e,p} - y_{e,p}|. \tag{14}$$

3. *Coefficient of Determination (R^2)*
   Another standard regression metric is the coefficient of determination (R^2). R^2 measures the proportion of variance in the true values $y_{e,p}$ that is explained by the model's predictions $\hat{y}_{e,p}$, providing a standardized evaluation of predictive accuracy. R^2 is defined as:

$$R_{t,p}^2 = 1 - \frac{\sum_{e \in E_t} (y_{e,p} - \hat{y}_{e,p})^2}{\sum_{e \in E_t} (y_{e,p} - \overline{y}_{e,p})^2}, \tag{15}$$

   where $\overline{y}_{e,p}$ is the mean of the observed values $y_{e,p}$. An R^2 value of 1 indicates a perfect fit, while an R^2 of 0 suggests that the model performs no better than simply predicting the mean $\bar{y}_{t,p}$. Negative values imply that the model introduces more error than a naive mean-based predictor.

   Unlike MSE and MAE, R^2 offers a relative measure of fit, which can be useful for comparing the prediction of the model on different street types with different variance. However, it can be sensitive to outliers and may be less reliable for highly skewed data distributions.

4. *Correlation Metrics: Pearson and Spearman*
   To further assess the model's performance, we compute the Pearson and Spearman correlation coefficients. Pearson correlation measures the strength of the linear relationship between actual and predicted values, indicating how well the predictions align in a linear sense. Spearman correlation, in contrast, evaluates the rank-based relationship, making it more robust to non-linear patterns. These correlation metrics complement MSE, MAE, and R^2, offering additional insights into the quality of the predictions.

5.2. Benchmarking Model Performance with Naive Baselines

Since no established baseline exists for this problem, we introduce a simple *mean-based baseline* to assess model performance. This baseline assumes that the predicted change in traffic volume for each edge is equal to the average observed change in traffic volume across all edges in the test set.

To evaluate model improvement over this baseline, we compare it across different metrics. For R^2, the baseline value is always 0, as a mean-based predictor does not explain any variance beyond predicting the average. For *Pearson and Spearman correlation*, the naive baseline assumes no predictive power, meaning that model predictions have no meaningful monotonic or linear relationship with the true values. As a result, the expected baseline values for both correlation coefficients are also 0.

While R^2 and correlation metrics are naturally bounded between 0 and 1, MAE does not have an inherent scale for comparison. To address this, we introduce a *normalized MAE* that expresses the error relative to the naive baseline. This ensures comparability across different evaluation metrics and allows for an intuitive interpretation of model performance:

$$1 - \frac{\text{MAE}}{\text{Naive MAE}}. \tag{16}$$

This transformation ensures that the normalized MAE ranges from 0 to 1, where 1 represents perfect predictions (zero error), and 0 indicates no improvement over the naive mean-based predictor. Applying the same normalization to *MSE* would result in a formulation that is mathematically identical to R^2 (Equation 15). Therefore, we do not report a normalized MSE separately.

By benchmarking against these naive baselines, we provide a clear and interpretable assessment of how much our surrogate model improves over a simple mean-based prediction.

5.3. The Impact of Simulation Stochasticity on Surrogate Model Accuracy

Agent-based models inhibit randomness. When running the same simulation under identical conditions, but with different random seeds, the outputs vary due to the probabilistic nature of agent-based interactions. Further, machine learning surrogates for ABMs rely on training data from downsampled populations, as using the full population is impractical due to the extensive computational resources required, increasing stochastic variability in the simulation outputs.

Stochastic variability acts as noise and affects the accuracy of surrogate models: In ML terms, this is called *aleatoric uncertainty*. Unlike *epistemic uncertainty*, which stems from incomplete knowledge and can be reduced with more data or improved modeling, aleatoric uncertainty reflects variability that cannot be eliminated with the given data, only quantified.

Given an optimal function F^* as stated in Equation 8, we seek to understand which role the aleatoric uncertainty, that stems from the agent-based simulation stochasticity, plays in the MSE.

$$\begin{aligned}
\text{MSE}(\hat{y}_{e,p}, y_{e,p}) &= \mathbb{E}\left[(y_{e,p} - \hat{y}_{e,p})^2\right] && (17)\\
&= \mathbb{E}\left[(y_{e,p} - \overline{y}_{e,p} + \overline{y}_{e,p} - \mathbb{E}[\hat{y}_{e,p}] + \mathbb{E}[\hat{y}_{e,p}] - \hat{y}_{e,p})^2\right] && (18)\\
&= \mathbb{E}\left[(y_{e,p} - \overline{y}_{e,p})^2 + 2(y_{e,p} - \overline{y}_{e,p})(\overline{y}_{e,p} - \mathbb{E}[\hat{y}_{e,p}])\right. && (19)\\
&\quad + 2(y_{e,p} - \overline{y}_{e,p})(\mathbb{E}[\hat{y}_{e,p}] - \hat{y}_{e,p}) && (20)\\
&\quad + (\overline{y}_{e,p} - \mathbb{E}[\hat{y}_{e,p}])^2 + 2(\overline{y}_{e,p} - \mathbb{E}[\hat{y}_{e,p}])(\mathbb{E}[\hat{y}_{e,p}] - \hat{y}_{e,p}) && (21)\\
&\quad \left. + (\mathbb{E}[\hat{y}_{e,p}] - \hat{y}_{e,p})^2\right]. && (22)
\end{aligned}$$

$\mathbb{E}[y_{e,p} - \overline{y}_{e,p}] = 0$ by definition of expectation. Further, $\mathbb{E}[(y_{e,p} - \overline{y}_{e,p})(\mathbb{E}[\hat{y}_{e,p}] - \hat{y}_{e,p})] = 0$,

since $y_{e,p} - \overline{y}_{e,p}$, representing the error due to randomness in the simulation, is independent of $\hat{y}_{e,p}$. The independence arises because $\hat{y}_{e,p}$ is learned solely from $\overline{y}_{e,p}$. Thus, the cross terms vanish, leaving:

$$\mathbb{E}\left[(y_{e,p} - \hat{y}_{e,p})^2\right] = \underbrace{\mathbb{E}\left[(y_{e,p} - \overline{y}_{e,p})^2\right]}_{\text{Aleatoric Variance}} + \underbrace{(\overline{y}_{e,p} - \mathbb{E}[\hat{y}_{e,p}])^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\mathbb{E}[\hat{y}_{e,p}] - \hat{y}_{e,p})^2]}_{\text{Surrogate Model Variance}}. \tag{23}$$

Each term represents a distinct source of error; while bias and variance measure how well the surrogate model fits the data (too much bias leads to underfitting, while too much variance leads to overfitting), the aleatoric variance represents the source of error that cannot be mitigated by improving the model architecture or training process.

Given a policy intervention p and $|R|$ independent simulation runs with different random seeds, the aleatoric variance quantifies the stochastic fluctuations in traffic volume for each edge e:

$$\sigma_{e,p}^2 = \frac{1}{|R|} \sum_{r=1}^{R} (y_{e,p}^r - \overline{y}_{e,p})^2, \tag{24}$$

where $y_{e,p}^r$ is the simulated traffic volume change on edge e in run r, and $\overline{y}_{e,p}$ is its mean across all runs. Computing $\sigma_{e,p}^2$ requires multiple simulations per scenario, which is computationally prohibitive. Now, recall the variance in the base case (Equation 2):

$$\sigma_e^2 = \frac{1}{|R|} \sum_{r=1}^{R} (v_e^r - \overline{v}_e)^2,$$

These computations have already been performed, as multiple runs have been performed for accurately estimating the traffic volumes in the base case.

Assumption. *For moderately restrictive policies, edge-level variance remains largely unaffected. Thus, one can approximate the variance of an edge e under policy p using the base case variance:* $\sigma_{e,p}^2 \approx \sigma_{e,b}^2$.

5.4. Estimating the impact of simulation stochasticity on the MSE

Model's performance is bounded by the aleatoric uncertainty. Hence, when interpreting the results, it is important to estimate the impact of the simulation stochasticity in the model performance. If our Assumption in Section 5.3 holds, it follows straight away from Equation 23:

$$\text{MSE}(F^*(x_e)) \geq \sigma_{e,b}^2 \tag{25}$$

Thus, the aleatoric component of the surrogate model's error could be *pre-computed* using the base case variance, allowing to establish a theoretical lower bound on MSE due to simulation stochasticity alone. Then, one could assess the impact on the surrogate model and determine how much of the model error is fundamentally unavoidable, depending on the factor of downsampling of the simulation, which directly impacts the simulation stochasticity, used in training data.

This would mark a key distinction for machine learning surrogates of ABM: unlike many machine learning applications where input uncertainty is difficult to quantify, here the source of aleatoric uncertainty is well-defined. We examine how simulation stochasticity, quantified as $\sigma_{e,b}^2$, relates to model error in our case study, as discussed in Section 7.4.

The irreducible noise in Equation 23 arises because the expected value $\overline{y}_{e,p} = \frac{1}{|R|} \sum_R y_{e,p}^r$ is typically not computed due to the high computational cost of running each policy multiple times. Instead, the base case variance $\sigma_{e,b}^2$ is readily available, as multiple simulation runs with different random seeds are already required as input for the model.

If $\overline{y}_{e,p}$ were available, the mean squared error $\text{MSE}(\hat{y}_{e,p}, \overline{y}_{e,p})$ would be:

$$\text{MSE}(\hat{y}_{e,p}, \overline{y}_{e,p}) = \mathbb{E}\left[(\overline{y}_{e,p} - \hat{y}_{e,p})^2\right] \tag{26}$$

$$= \underbrace{(\overline{y}_{e,p} - \mathbb{E}[\hat{y}_{e,p}])^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{y}_{e,p} - \mathbb{E}[\hat{y}_{e,p}])^2]}_{\text{Surrogate Model Variance}}. \tag{27}$$

Thus, if $\overline{y}_{e,p}$ were available for every policy intervention p, the aleatoric variance would be zero. A more in-depth analysis of this phenomen can be found in Appendix A.

6. Case Study

To evaluate the effectiveness of our surrogate model, we test it on the large-scale MATSim model of Paris, France. The model is based on a synthetic population of the Île-de-France region surrounding Paris (Hörl and Balac, 2021) which describes households, persons with their sociodemographic attributes and their daily activity sequences including information on when and where (by coordinate) the activities (home, work, shopping, ...) are performed. The process is open-source and entirely based on open data, which makes it fully replicable. In our case, 1% of households from the population of 12 million people are selected for performance reasons. The downsampled synthetic population is then simulated using MATSim for a baseline case of the entire region around Paris, and then cut so that only the agents

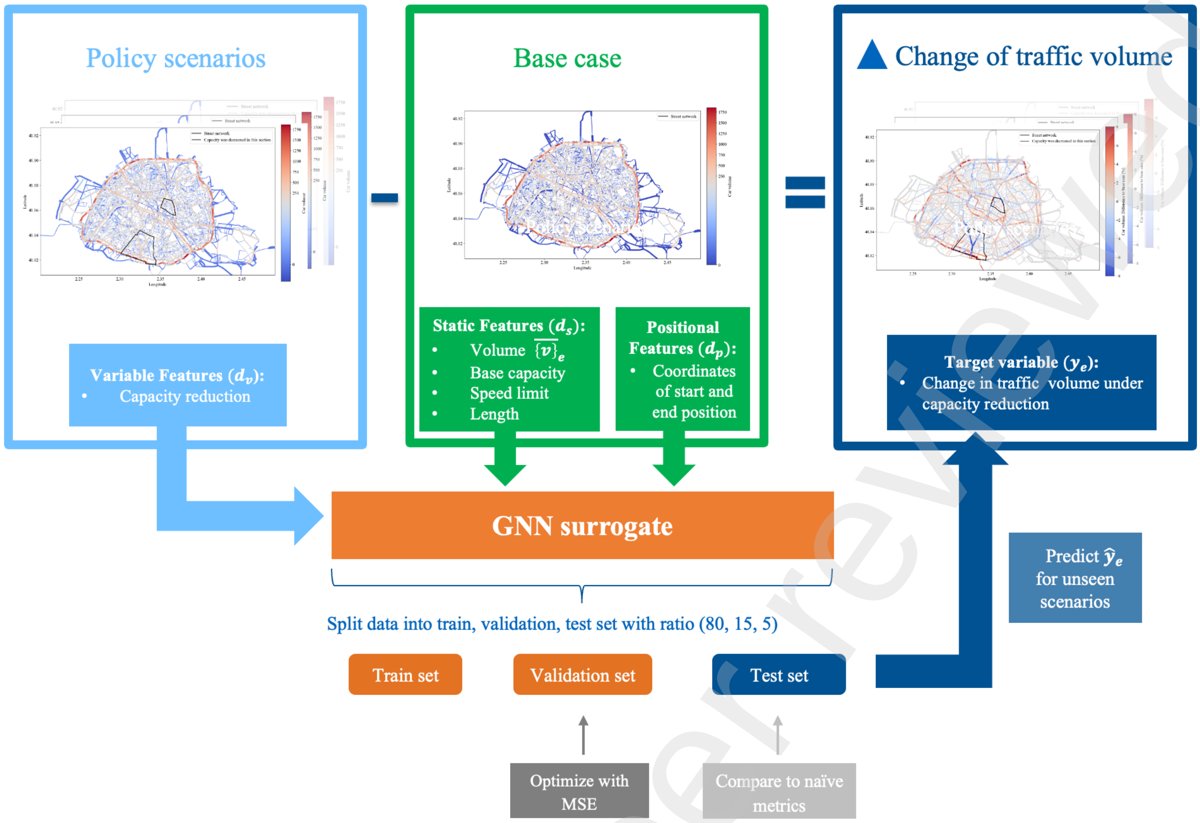


Figure 3 – *Workflow of the case study*

interacting with the city area and its surrounding high-capacity ring street (*boulevard périphérique*) are retained. The MATSim model simulates a typical workday, and we aggregate the traffic volumes at the link level over the entire day. As a result, traffic volume is expressed in units of vehicles per day (veh/day).

Figure 3 provides an overview of the GNN surrogate model designed to predict traffic volume changes resulting from capacity reduction policies. The process begins with link-level inputs: *policy scenarios*, simulated in MATSim, corresponding to combinations of districts where capacity reductions have been applied (see Section 6.1); the *base case*, representing the network without interventions and providing static features such as baseline volume, capacity, speed limits, and link length and *positional features*, capturing the start and end coordinates of each link (see Section 3.1). For each simulated policy scenario, we compute the difference in traffic volume at the link level between the simulated policy scenario and the base case, establishing the target variable representing traffic volume changes due to the policy intervention. The GNN surrogate model is trained on these inputs, with the dataset split into training, validation, and test sets. It optimizes performance using the MSE loss function.

We first explain the simulation setup, then the data preparation, and finally, the training process of the surrogate model.

6.1. Simulation Setup

We begin by generating 10,000 scenarios, each of which has the policy "50% Capacity Reduction" (p) being applied to different combinations of districts. The tested policy p enforces a 50% capacity reduction on all primary, secondary, and tertiary roads within the selected districts in a scenario. The street network consists of approximately 35,000 street segments, with primary, secondary, and tertiary roads (as classified by OpenStreetMap) constituting about 15,000 of these.

Parallelly, we compute the MATSim base case (see Section 3.1) by averaging the traffic volume per street segment across 50 independent runs with different random seeds. The base case serves as a reference point for quantifying the changes in traffic volume, due to the introduced policy.

Paris is divided into 20 districts ("Arrondissements"). To generate district combinations, we follow a two-step process. First, we determine the size of each combination, i.e., the number of districts included.

If chosen randomly, there are

$$\sum_{i=0}^{20} \binom{20}{i} = 1,048,576$$

possible combinations. The selection of district combinations follows a binomial distribution, $X \sim \text{Binomial}(20, 0.5)$, with a mean of 10 and a standard deviation of approximately 2.24.

To ensure the model accurately captures the localized effects of policy interventions, we focus on a smaller set of districts. Inspired by real-world transformations in Paris, where efforts have prioritized active mobility by expanding bike lanes and reallocating street space (Natterer et al., 2024), capacity reduction policies have been primarily implemented in Arrondissements 1 - 4, representing the "core" of the city. Therefore, we aim to generate district combinations with around four districts. To construct a diverse dataset, we generate 10,000 combinations using two different sampling strategies:

1. Randomly sample 5,000 combinations with a set length following a normal distribution with mean 4 and standard deviation 1.
2. Randomly sample 5,000 combinations with a set length following an exponential distribution with mean 4.

We run simulations for the selected combinations, resulting in 8,308 successfully completed runs. Each simulation took approximately one hour to complete on a high-performance cluster using 12 cores. The completed simulations are then used for training and testing the surrogate model.

6.2. Data prepration

The resulting simulations are first made compatible with the GNN by turning it into the P_{sim} format (as described in Section 4.1). The corresponding dual graph has $31,635$ street segments (nodes), and $59,851$ edges (representing intersections). As illustrated in the top half of Figure 2, to analyze policy effects, we subtract the base case traffic volume from the intervention case (resulting from policy simulations), generating $y^r_{e,p}$, which acts as the ground truth for the GNN surrogate. At the network level, $y^r_{e,p}$ can be visualized as a difference plot, highlighting changes in traffic volume. Red areas indicate an increase in traffic volume compared to the base case, while blue areas represent a decrease, both driven by the policy intervention. Results presented in this format are shown in Figures 8, 9, and 10.

As illustrated in Figures 2 and 3, each street segment is characterized by three types of features: static, variable, and positional. The selected features include:

1. *Static features (d_s)*:
   - Traffic volume base case ($\overline{v}_e$)
   - Capacity base case
   - Speed limit
   - Length
2. *Variable features (d_v)*:
   - Capacity reduction
3. *Positional features (d_p)*:
   - x and y coordinates of start and end position

Therefore, in our setup, $d_s = 4$, $d_v = 1$, and $d_p = 2$, resulting in an input feature dimension of 7 for the first PointNet convolution layer in the GNN surrogate. Initially, we considered a broader set of features, including categorical attributes such as street types and allowed transport modes. However, ablation tests revealed that the model did not effectively learn from these additional features, leading us to focus on the selected subset. For the second PointNet convolution layer, the input dimension equals 514 (incorporating end position). See Section 4.2 for more details.

As a preprocessing step, all the features, being continuous, are normalized using a standard scaler, ensuring they have zero mean and unit variance. This helps maintain numerical consistency across features and improves model convergence. Finally, the processed data is split into training, testing, and validation sets with a ratio of (80, 15, 5), yielding 6,646 scenarios for training, 1,246 for validation, and 416 for testing.

6.3. Training process of the surrogate model

The GNN surrogate model is trained to predict changes in traffic volume on each street segment resulting from a policy intervention, relative to the base case. Its generalization ability is evaluated on unseen district combinations. The model is optimized using Mean Squared Error (MSE), ensuring accurate predictions while capturing network-wide traffic dynamics. As a baseline for performance, we use the Naïve MSE. Additionally, we log Spearman and Pearson correlation metrics to assess the consistency of the predictions.

The GNN-Surrogate was implemented in PyTorch Geometric and trained for 750 epochs, with early stopping triggered if validation performance did not improve for 40 consecutive epochs (approximately 5% of total training). To stabilize learning, a linear warm-up phase was applied for the first 5% of the total epochs (~ 37.5 epochs), gradually increasing the learning rate before reaching the peak value of 5×10^{-4}. The training process then followed a cosine decay learning rate schedule, progressively reducing the learning rate to 5×10^{-6} by the end to enhance convergence. Gradient clipping was applied to prevent exploding gradients. The optimizer used was AdamW, with a weight decay of 1×10^{-4}, acting as a regularizer to mitigate overfitting.

During the training process, the model's performance was evaluated using the validation loss, R^2, and correlation metrics (Pearson and Spearman coefficients). The learning curves are illustrated in Figures 4 and 5. The first figure shows that both training and validation loss (red and blue, respectively) decrease over time, while R^2 (green) increases, indicating improved predictive accuracy. Training stabilizes after approximately 700 epochs, with validation loss plateauing and correlation metrics reaching their peak. The second figure highlights the steady increase of both Pearson (brown) and Spearman (purple) correlations throughout training, demonstrating that the model progressively learns meaningful relationships in the data.

Training was conducted on a single NVIDIA H100 GPU, taking approximately 23 hours to converge. After training, a single forward pass through the GNN takes less than a second, compared to the 1-hour computation time required to obtain results from the (downsampled) simulation.

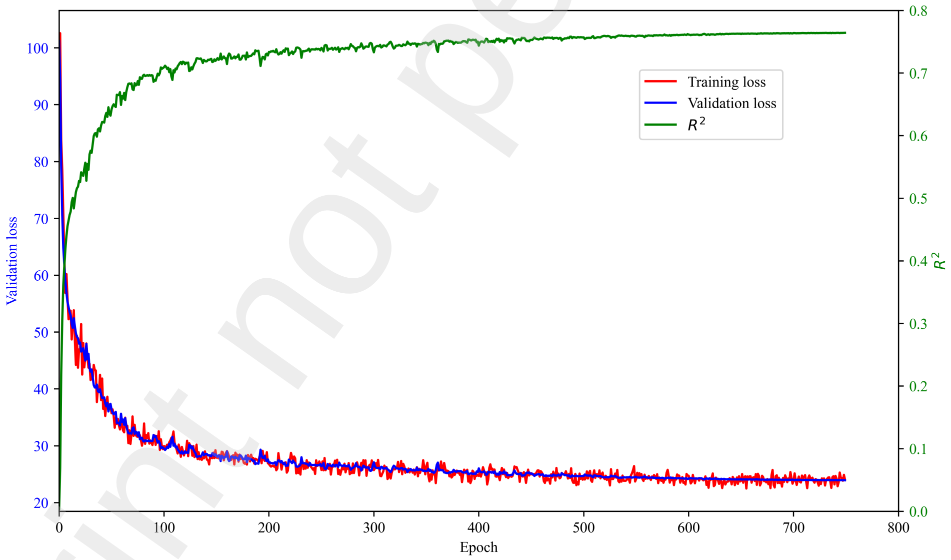


Figure 4 – *Validation loss, training loss and* R^2

7. Results

7.1. Overall performance

The trained surrogate model can predict the effects of policy interventions in randomly selected district combinations within milliseconds. It achieves an overall accuracy of $R^2 = 0.76$ and a MSE of 24.95 on the test set, significantly outperforming a naive baseline prediction (Naive MSE = 103.47). The base case variance is 8.88, which would serve as an estimation for a lower bound for MSE, representing the aleatoric variance - if Assumption 5.3 holds. The model achieves a MAE of 2.74 (Naive MAE = 3.96),

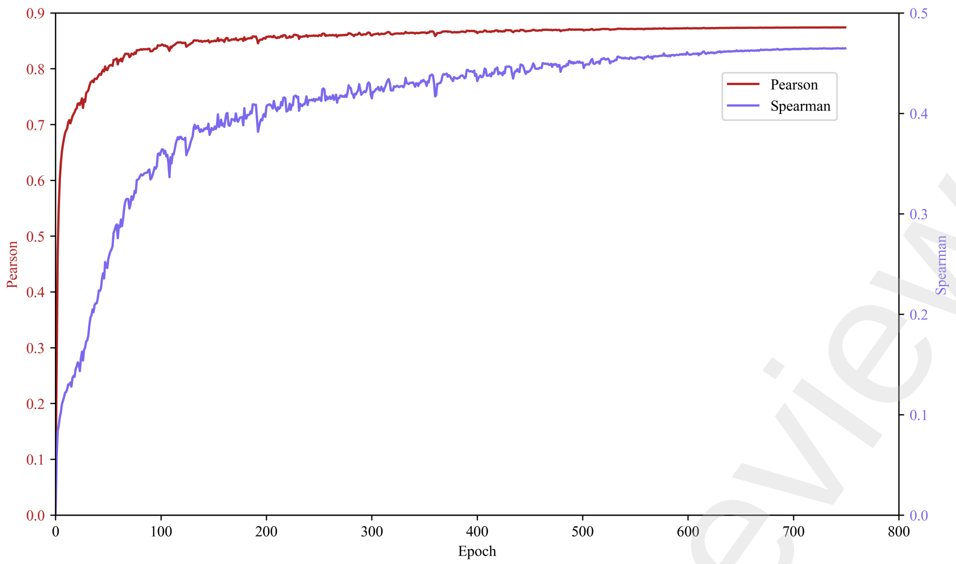


Figure 5 – *Spearman and Pearson Correlation over the training process*

and strong predictive correlations, with a Spearman correlation of 0.47 and a Pearson correlation of 0.87.

In this section, we first evaluate the model's performance across different metrics and street types (Section 7.2), with a particular focus on the differences in predictions for roads with and without capacity reductions (Section 7.3). Further, we investigate the relationship between simulation stochasticity and model error in Section 7.4, as motivated in Section 5.3. Section 7.5 analyzes performance consistency across multiple evaluation metrics. Finally, we analyze results at the network level (Section 7.6): we first present results for three randomly selected test scenarios to provide a general overview, then we overlay results from all test scenarios to assess the overall error distribution across the network.

7.2. Overall Performance Across Street Types

To evaluate the model's performance across the test set, we analyze key aggregated performance metrics for different street types, as shown in Table 3. The table presents for each street type, the number of roads that it consists of (Count), the average traffic volume counted throughout a day in the base case $\overline{v}_t$ (as calculated in Section 3.1), the average variance in the base case $\sigma^2_{t,b}$ (Eq. 2), the variance across street types σ^2_t (Eq. 12), the MSE per street type (Eq. 13), the MAE per street type (Eq. 14) and R^2 (Eq. 15). The table is sorted by its traffic volume in the base case $\overline{v}_t$. Since only a single policy p was tested, the subscript p is omitted in this section for clarity.

Street Type t	**Count**	$\overline{v}_t$	$\sigma^2_{t,b}$	σ^2_t	$\mathbf{MSE}_t$	$\mathbf{MAE}_t$	$\mathbf{R^2}_t$
All Roads	31,635	52.21	8.88	103.42	24.95	2.74	0.76
Trunk	1,011	**505.16**	18.87	209.05	105.28	6.22	0.51
Primary	6,112	117.79	20.23	405.80	59.11	4.87	**0.86**
Secondary	4,715	51.99	13.13	82.17	29.10	3.52	0.65
Tertiary	4,130	36.54	11.05	47.47	19.87	3.00	0.58
Residential	12,474	13.49	4.44	16.88	11.88	2.03	0.31
Living Streets	774	4.82	**1.20**	**4.40**	**3.67**	**1.18**	0.17

Table 3 – *Model Performance Metrics by Street Type (aggregated over the test set, as defined in Section 5.1)*

The street network consists of approximately 35,000 street segments, with primary, secondary, and tertiary roads constituting about 15,000 of these. Residential roads make up the largest category (nearly 13,000 roads), whereas living streets are the least common, with only 774 street segments. The policy of "50% Capacity Reduction" has been implemented in all primary, secondary, and tertiary roads if in the relevant scenario. The average traffic volume varies significantly depending on the street type. Trunk roads exhibit the highest base volume, with an average of 505.16 vehicles per day, while living streets have the lowest, averaging just 4.82 vehicles per day. We make the following key observations:

1. *Higher traffic volume in the base case leads to greater model accuracy*: Across all street types, higher base case traffic volumes are strongly associated with higher absolute errors in terms of MSE and MAE.

2. $\sigma_{t,b}^2$ and σ_t^2 are highly correlated, with a Pearson correlation of 0.85. They do, however, measure different aspects of variability:
   - $\sigma_{t,b}^2$ quantifies variance within simulation runs across multiple random seeds for the base case.
   - σ_t^2 captures variance across all roads of a given type, reflecting differences in traffic volume changes between roads.
3. *$\sigma_{t,b}^2$ and σ_t^2 generally correlate with higher base traffic volume and accuracy, except for trunk roads*: For most street types, greater traffic volume corresponds to higher variance across multiple simulation runs and across all streets of a given type, leading to improved predictive accuracy. However, trunk roads deviate from this trend.

4. *Trunk roads form an exception to the variance-accuracy relationship*: Despite having the highest base traffic volume - approximately five times that of primary roads - trunk roads exhibit lower variance ($\sigma_{t,b}^2$, and σ_t^2) than primary roads.
   - $\sigma_{t,b}^2$ being lower for trunk than for primary roads means that trunk roads remain stable across different simulation runs with random seeds. For example, in Paris, the motorway ring street around the city, the Boulevard Périphérique, consists of trunk street segments. Since this street is less affected by route-choice variability compared to other roads, such as primary roads, traffic volume remains relatively consistent, with minimal stochastic fluctuations during simulation runs.
   - The relatively low σ_t^2 for trunk roads suggests that trunk roads indicates that traffic patterns among trunk roads are more uniform compared to primary roads. This suggests that individual trunk roads exhibit similar traffic behavior, whereas primary roads show greater variability.
   - Additionally, the low σ_t^2 contributes to the relatively low R^2. Since R^2 measures the proportion of variance explained by the model, lower variance inherently results in lower R^2 values - even for roads with high traffic volumes. The phenomen of lower variance leading to lower R^2 values, and a seeming contradiction between absolute error metrics (MSE/MAE) and R^2 is further explored in the next section (Section 7.3), where we examine the distinction between roads with and without capacity reduction.
5. *Poor performance on low-volume roads*: In terms of R^2, the model performs worst on roads with comparatively smaller traffic volumes, such as residential ($R^2 = 0.31$) and living streets ($R^2 = 0.17$), indicating that these street types exhibit the least predictable traffic patterns.

7.3. Impact of Capacity Reduction on Model Performance

Table 4 presents the model's performance across different street types, distinguishing between roads with and without capacity reduction. Since reductions apply only to Primary, Secondary, and Tertiary roads, the analysis is limited to these categories.

The number of observations (*Count*) represents the total street segments evaluated across all test scenarios. For instance, the 594,435 observations for primary roads with capacity reduction indicate the total number of primary street segments affected at least once across scenarios. Overall, the dataset includes 1.4 million street segments with capacity reduction and 4.3 million without.

A consistent pattern emerges across all street types: roads with capacity reduction exhibit higher R^2, Pearson, and Spearman correlation values, but also show higher MSE and MAE. This apparent contradiction between error metrics warrants further investigation. Looking closer, we observe also patterns in the characteristics of the two types:

1. Roads with capacity reduction exhibit significantly higher variance, with primary roads, for example, showing a variance of 1,250.88 compared to just 90.63 for those without. Across all categories, variance for roads with capacity reduction is approximately eleven times greater than for roads

Street Type t	Count	$\sigma^2_{t,b}$	σ^2_t	MSE$_t$	MAE$_t$	R$^2{}_t$	Pearson	Sp.
P/S/T, Cap. Red.	1,444,212	86.03	626.97	42.01	4.54	**0.93**	**0.91**	**0.82**
P/S/T, No Cap. Red.	4,349,420	70.58	56.20	**35.65**	**3.58**	0.37	0.58	0.51
Primary, Cap. Red.	594,435	134.29	1250.88	61.28	5.47	**0.95**	**0.94**	**0.85**
Primary, No Cap. Red.	1,721,021	112.78	90.63	**55.09**	**4.45**	0.39	0.61	0.54
Secondary, Cap. Red.	430,234	63.16	208.84	32.42	4.21	**0.85**	**0.90**	**0.82**
Secondary, No Cap. Red.	1,423,878	49.35	42.01	**27.59**	**3.24**	0.34	0.58	0.51
Tertiary, Cap. Red.	419,543	41.12	107.42	24.55	3.56	**0.77**	**0.87**	**0.79**
Tertiary, No Cap. Red.	1,204,521	35.38	23.66	**17.42**	**2.73**	0.26	0.54	0.48

- "P/S/T" stands for "Primary, Secondary, and Tertiary" roads.
- "Cap. Red." refers to roads affected by capacity reduction policies, while "No Cap. Red." refers to roads unaffected by these policies across the test set.
- "Pearson" denotes the Pearson correlation, and "Sp." refers to the Spearman correlation.

Table 4 – *Model Performance Metrics by Street Type, with and without Capacity Reduction*

without. This increased variance naturally inflates R^2 values, as variance in the target variable directly affects its computation:

$$R^2_t = 1 - \frac{MSE_t}{\sigma^2_t}$$

The higher variance is expected, as roads with capacity reduction experience more extreme fluctuations in traffic across scenarios. Some street segments are *directly affected* in one scenario (where their capacity is reduced) but only *indirectly influenced* in another (where no reduction occurs). This alternation between scenarios leads to a wider spread in observed traffic volume changes, explaining the high variance. Consequently, while R^2 may indicate strong model performance, it can be misleading when variance is high, as absolute errors remain significant despite the improved relative fit.

2. Roads with capacity reduction systematically exhibit higher base case variance ($\sigma^2_{t,b}$) compared to those without. Investigating this, we find a difference arises from a sampling bias, as detailed in Appendix B. Specifically, our selection method tends to choose districts with higher base case traffic volumes, which in turn have higher variance. Thus, the observed base case variance differences stem from a sampling artifact rather than a causal effect of capacity reduction, influenced by district selection frequencies and their base volumes.

The first observation explains the high R^2 values observed for roads with capacity reduction. Together with the Pearson and Spearman correlations, these results confirm that the model performs better on roads with capacity reduction than on those without. Furthermore, if Assumption 5.3 holds, then the higher base case variance should also result in a higher MSE (see Section 5.4). This aligns with our findings and provides a clear explanation for the observed differences. Consequently, we conclude that the elevated MAE and MSE values for roads with capacity reduction are a direct consequence of sampling bias and increased base case variance.

Our analysis further shows that base case variance accounts for a stable proportion of MSE across different policy scenarios, reinforcing Assumption 5.3. This suggests that stochasticity in the base scenario remains a strong predictor of variance under capacity reduction policies. However, to fully validate this assumption, we must empirically examine whether edge-level variance remains stable across policy interventions. This leads to the next section, where we investigate the relationship between simulation stochasticity and model error.

7.4. Relationship of Simulation Stochasticity and Model Error

In Section 5.3, we introduced an assumption, which states that for moderately restrictive policies, edge-level variance remains largely unaffected, suggesting to approximate the variance under policy p using the base case variance: $\sigma^2_{e,b} \approx \sigma^2_e$. Here, we empirically investigate the validity of this assumption.

7.4.1. Implications for Model Error

If the assumption in Section 5.3 holds, we would expect a strong relationship between base case variance and model error across different scenarios.

Street Type t	$\sigma^2_{t,b}$	$\mathbf{MSE}_t$	**Share of** $\sigma^2_{t,b}$ **in** $\mathbf{MSE}_t$
All Roads	8.88	23.35	38.03%
Trunk	18.87	85.78	22.00%
Primary	20.23	58.56	34.55%
Primary, Cap. Red.	24.09	72.98	33.01%
Primary, No Cap. Red.	19.24	54.35	35.40%
Secondary	13.13	27.54	47.69%
Secondary, Cap. Red.	16.11	33.99	47.39%
Secondary, No Cap. Red.	12.42	25.89	47.96%
Tertiary	11.05	18.92	58.43%
Tertiary, Cap. Red.	12.50	24.54	50.95%
Tertiary, No Cap. Red.	10.70	17.38	61.58%
Residential	4.44	10.65	41.69%
Living Streets	1.20	3.34	35.93%

Table 5 – *Base Case Variance and it's contribution to MSE*

Table 5 examines the contribution of base case variance to MSE across street types. The proportion of MSE explained by base case variance is highest for tertiary roads without capacity reduction (61.58%) and lowest for trunk roads (22.00%). This suggests that for tertiary roads, most of the model's error is attributable to inherent traffic variability, whereas for trunk roads, a larger portion of the error stems from other factors, such as complex traffic interactions or model limitations. Note that if this proportion were 100%, provided that the assumption holds, the MSE could be entirely explained by base case variance, meaning no further model improvements would be possible. Thus, the surrogate model is often performing close to its theoretical limit, constrained by the uncertainty in the training data.

Generally, secondary and tertiary roads have higher proportions of base case variance in MSE, while trunk and primary roads with capacity reduction have lower shares, suggesting greater prediction challenges in these areas.

An important observation is that within each street type, the share of base case variance in MSE remains relatively stable, regardless of whether capacity reduction is applied. For example, primary roads have a share of 33.01% with capacity reduction and 35.40% without, while secondary roads remain around 47.39% and 47.96%, respectively. However, tertiary roads show a larger difference (50.95% vs. 61.58%). This suggests that:

1. Base case variance ($\sigma^2_{t,b}$), derived from the base scenario alone, remains a strong predictor of stochasticity even when capacity reduction is introduced. This supports our assumption that edge-level variance remains largely stable under moderate policy interventions.
2. While absolute values of MSE and variance increase with capacity reduction, their ratio remains relatively stable. This indicates that increased prediction error (MSE) under capacity reduction is largely proportional to the inherent stochasticity of each street type rather than additional obstacles for the model.
3. The characteristic ratio of variance to MSE varies across street types but follows a predictable pattern - around 34% for primary roads, 48% for secondary roads, and 55% for tertiary roads. This suggests that each street type has an intrinsic relationship between inherent stochasticity and prediction error, though the degree of stability varies depending on the street type, and not so much on the policy intervention.

Figure 6 illustrates the relationship between model error and simulation stochasticity across different street types. Simulation stochasticity, measured as variance of the base case, is shown on the x-axis, while model error, represented by MSE, is displayed on the y-axis. Each dot corresponds to a street type, positioned according to its variance and MSE, with the blue dashed line representing the best-fitting regression line. The grey dashed line denotes the lower bound on the MSE, as established in Section 5.3.

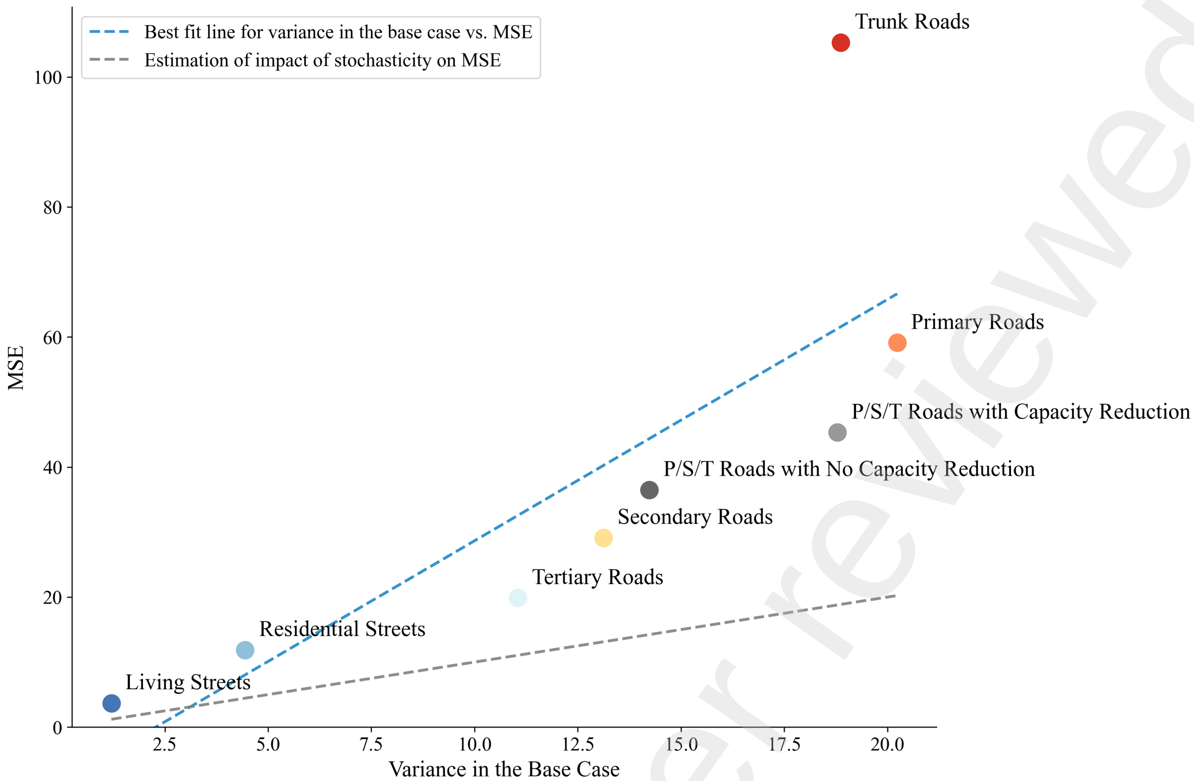


Figure 6 – *Relationship between base case variance and MSE across street types. The strong correlation (r = 0.80) suggests that base case variance is a reliable predictor of model error across different policy scenarios.*

The strong correlation between MSE and base case variance confirms that simulation stochasticity is a major driver of model error. Most street types align well with the expected relationship between the MSE and the simulation stochasticity, except for trunk roads, which exhibit disproportionately high MSE relative to their variance. This discrepancy is likely due to their higher traffic volumes but lower fluctuation compared to other street types.

We conclude that in our use case, the base case variance can serve as a reliable estimator for prediction uncertainty. This assumption holds also because policy interventions affect only a limited subset of roads in each scenario. While up to 14,947 roads (namely primary, secondary, and tertiary roads) of the 31,635 roads are eligible for capacity reduction, in practice, only about 3,000 roads (roughly 10% of the total network) are modified per scenario, assuming an even distribution across four districts on average. While some deviations are expected, our findings suggest that base case variance provides a reliable and easily obtainable approximation of aleatoric variance arising from simulation stochasticity in surrogate models for agent-based models.

7.5. Evaluation Across Multiple Metrics

Figure 7 presents a radar plot that visualizes model performance across different street categories using all presented evaluation metrics, sorted by their R^2 value. To facilitate a fair comparison across metrics, MAE values are normalized using their respective naive baseline values, constraining them between 0 and 1, where higher values indicate better performance (see Section 5.2). For clarity, the radar plot groups roads with capacity reduction and those without together. Each street type exhibits a distinctive polygonal trace, forming a diamond-like shape with relatively consistent scores across metrics. The radial grid lines at 0.2 intervals highlight substantial performance differences, with deviations of up to 0.3 - 0.4 between the best- and worst-performing categories.

The radar plot shows that the trends observed in Section 7.2 and Section 7.3, corresponding to Table 3 and Table 4, remain consistent across evaluation metrics:

1. *General Hierarchy of Street Types*: The ranking of street types in terms of accuracy remains consistent across multiple error metrics. Specifically, primary roads demonstrate the highest accuracy across all metrics, while Secondary roads achieve best in all but Spearman correlation. Tertiary roads rank next for R^2 and Pearson correlation, whereas Trunk roads outperform them in MAE

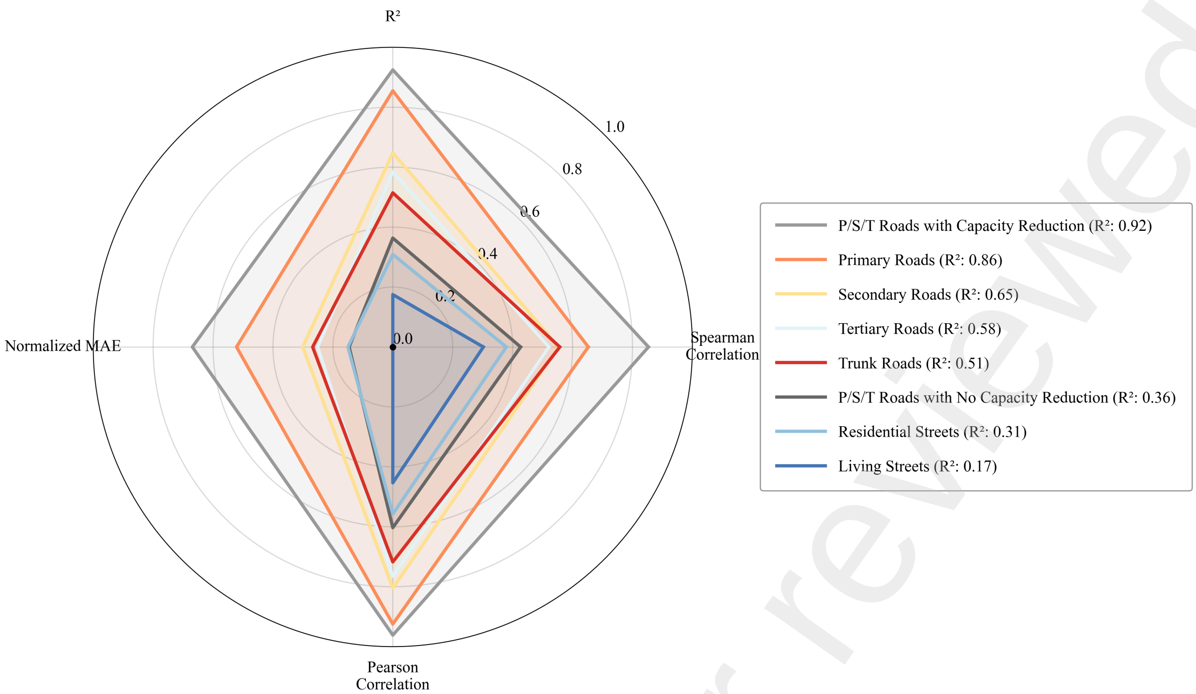


Figure 7 – *Radar plot comparing model performance metrics across different street types, sorted by R^2 value.*

and Spearman correlation. This ranking is consistently followed by residential streets, with living streets exhibiting the lowest performance across all metrics.

2. *Impact of Capacity Reductions:* The trend observed in Table 4 is also reflected across additional evaluation metrics: Primary, Secondary, and Tertiary roads with capacity reductions consistently outperform their counterparts without capacity reductions across all metrics.

Thus, the generally consistent diamond shapes across metrics suggest that the model's performance is stable across different evaluation criteria. This indicates that variations in accuracy stem from inherent differences between street types rather than inconsistencies in the evaluation metrics.

7.6. Network-Level Evaluation: Visual Analysis and Interpretation

To extend our analysis from link-level to network-level performance, we examine both specific test cases and aggregate patterns.

7.6.1. Individual Test Case Analysis

We first present three randomly selected test cases from our test set in Figures 8, 9, and 10. These figures compare the outputs of MATSim (top) with our surrogate model predictions (bottom), showing changes in traffic volume relative to the base case (see Section 3.1). The color scale ranges from blue (50% decrease) to red (50% increase) in traffic volume.

For Test Case 1, we observe a MSE of 2.21 and $R^2 = 0.57$, compared to a naive baseline MSE of 35.86. Test Cases 2 and 3, while showing higher MSE (4.16 and 3.68 respectively), achieve better R^2 scores (0.81 and 0.79) due to their substantially higher baseline MSE (152.91 and 125.33). This pattern indicates that while Test Cases 2 and 3 present more challenging prediction tasks with higher variance in the target variable, the model more effectively captures the underlying patterns relative to the mean prediction.

The model demonstrates strong performance in predicting traffic reductions due to capacity reduction policies, visible as blue segments in the plots. However, sometimes the model sometimes struggles to accurately represent evasive behavior: when comparing the top of the figure (MATSim output) with the bottom of the figure (surrogate model output), red street segments appear in the surrogate model's output that are absent in the MATSim results.

This suggests that the model sometimes falsely detects evasive behavior when none has actually occurred. Addressing this issue would likely require additional training data to improve the model's ability to differentiate between genuine evasive maneuvers and regular traffic patterns.

These findings align with our quantitative analysis showing generally higher accuracy for roads with capacity reduction and higher traffic volumes, across all evaluation metrics (Section 7.5).

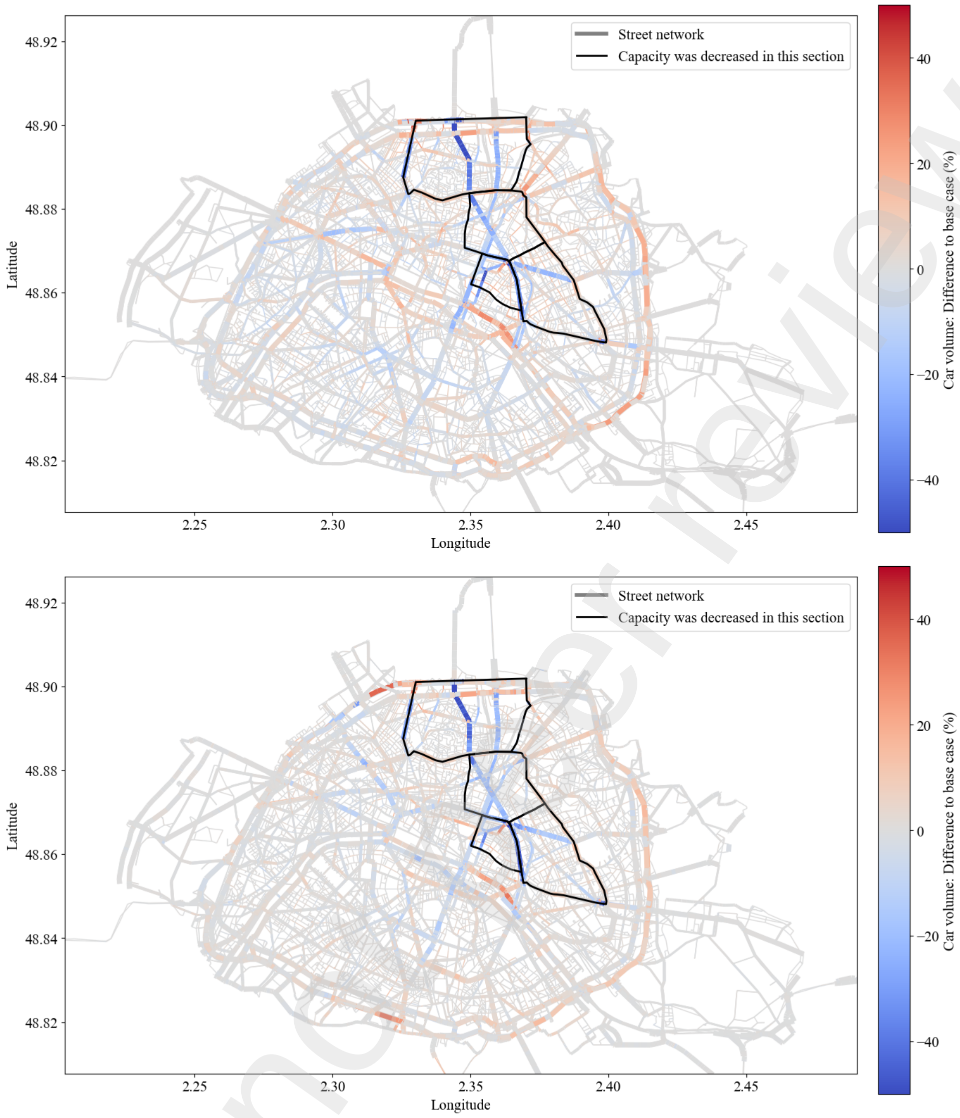


Figure 8 – *Test case 1: Simulated (top) vs. predicted (bottom) change in traffic volume.*

7.6.2. Aggregate Pattern Analysis

To assess the spatial distribution of prediction errors across Paris, we computed and visualized two complementary maps: one showing absolute differences and the other showing relative differences between predicted and actual traffic volumes (Figures 11 and 12). For computing this, we first computed difference plots for each test scenario by calculating the discrepancy between MATSim-simulated and model-predicted traffic volume changes at the edge level. As illustrated in Figures 8, 9, and 10, we then aggregated these differences across all test cases to compute the average prediction error δ_e for each edge $e \in E$, providing comprehensive assessment of model performance across the entire test set:

$$\delta_e = \sum_{s \in S} \frac{|y_{e,p,s} - \hat{y}_{e,p,s}|}{y_{e,p,s}} \tag{28}$$

The absolute differences map visualizes raw prediction errors in vehicle counts, with discrete thresholds at 5, 10, and 20 vehicles to highlight varying levels of prediction accuracy. Meanwhile, the relative differences map normalizes errors by the base traffic volume of each street segment, using 10%, 50%, and 100% thresholds to account for differences in traffic scale across various street types and locations.

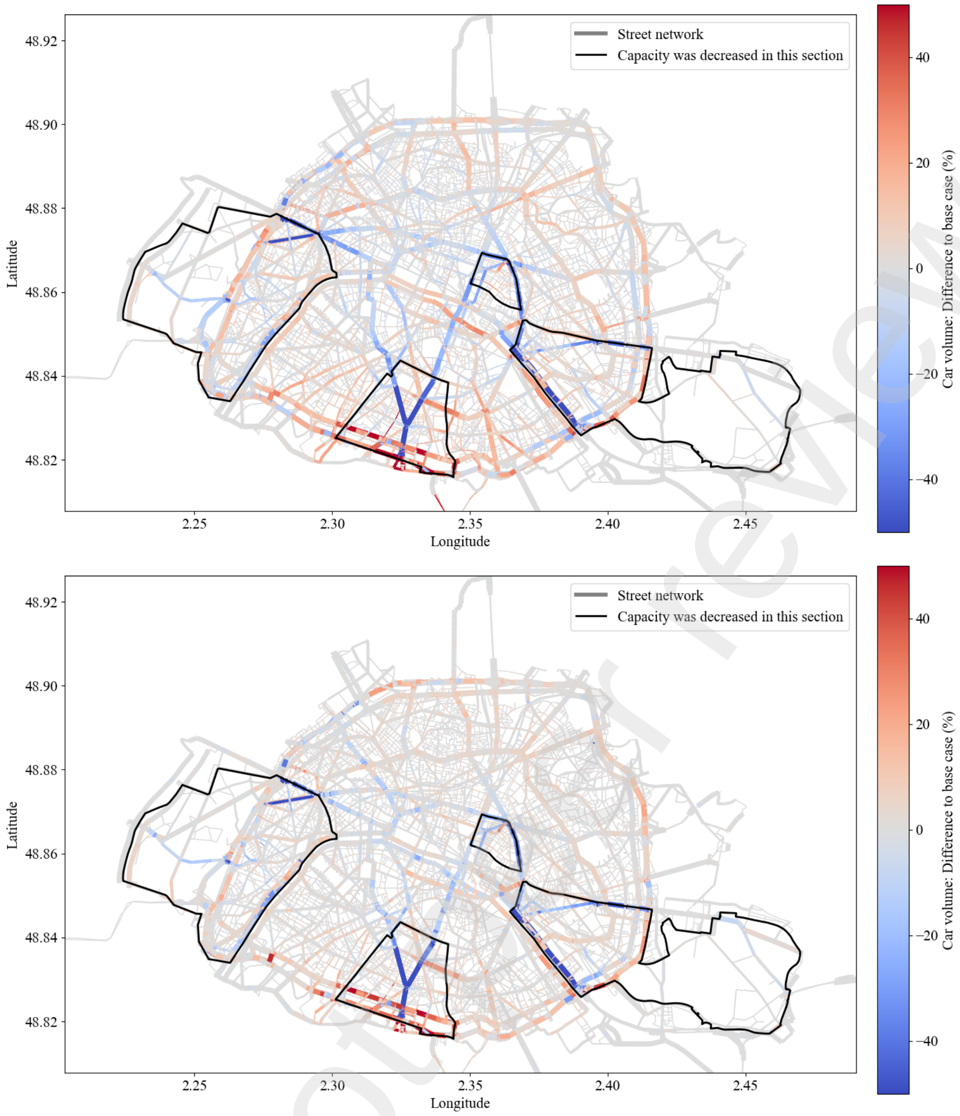


Figure 9 – *Test case 2: Simulated (top) vs. predicted (bottom) change in traffic volume.*

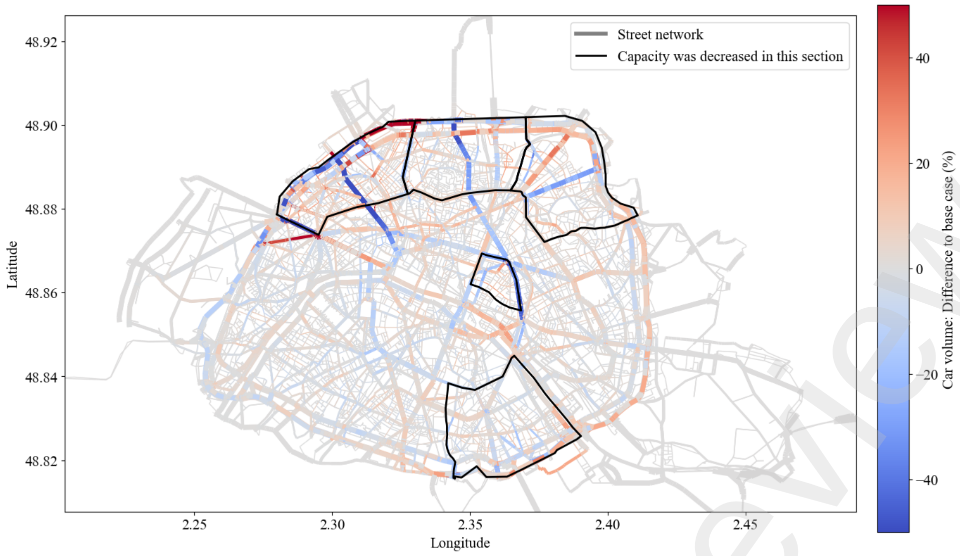
This dual visualization approach reveals an interesting spatial pattern: while absolute errors are higher in central areas (decreasing from 3.41 to 1.71 vehicles from center to periphery), relative errors show an inverse relationship, increasing from 20.2% in the center to 36.8% in peripheral areas. This pattern can be attributed to the higher and more stable traffic volumes in central areas compared to the more variable traffic patterns in peripheral zones, where even smaller absolute errors can translate into larger percentage deviations.

In both representations (relative and absolute), it is evident that the model achieves high accuracy on the main street network. This aligns with the results in Section 7.6, indicating that roads with higher traffic volumes show generally higher accuracy (across all metrics, see Section 7.5).

8. Conclusion

This study explores the use of machine learning as surrogate models for large-scale agent-based transport simulations. The applied graph neural network can succesfully learn the effects of capacity reduction policies on urban traffic patterns and captures changes in traffic volume on link level. In a case study using the MATSim model of Paris, the key findings were as follows:

- **Surrogate Model Performance:** The GNN-based model accurately predicts traffic changes due to capacity reduction policies, achieving an R^2 of 0.76 and strong correlation metrics on the overall



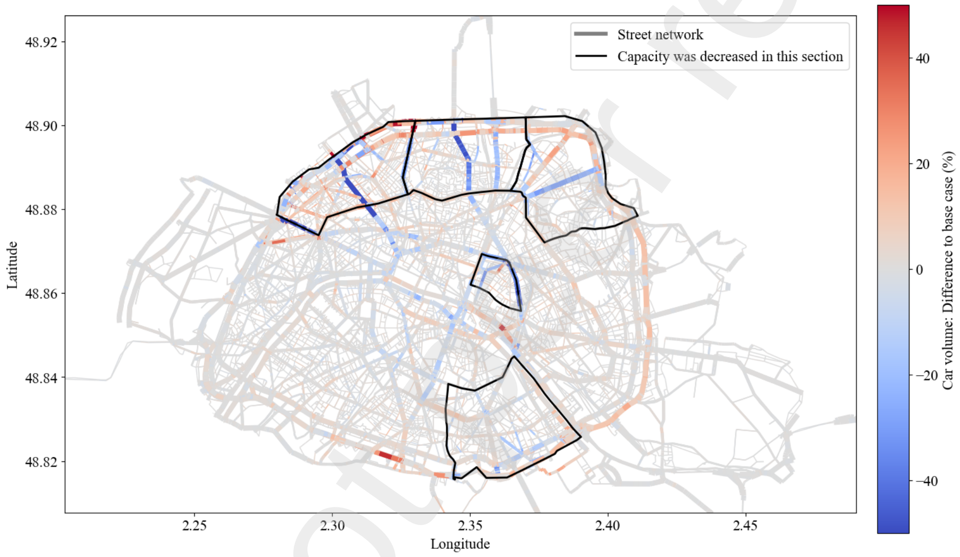


Figure 10 – *Test case 3: Simulated (top) vs. predicted (bottom) change in traffic volume.*

test set. While absolute errors remain, the model outperforms naive baselines and captures spatial dependencies effectively.

- **Variance and Model Error:** The variance in the base case proves to be a strong predictor of model error. The share of base case variance in the Mean Squared Error remains stable across street types, confirming that simulation stochasticity is a dominant factor influencing surrogate model errors.

8.1. Limitations and Future Work

Despite the strong performance of the surrogate model, several limitations remain that should be addressed in future research.

- *Computational Cost and Limited Transferability*
  While the trained GNN surrogate enables rapid policy evaluation, its initial training process is computationally intensive. In our case, 10,000 full-scale ABM simulations - each requiring approximately one hour on a high-performance computing cluster - were necessary before deployment. This substantial preprocessing effort limits adaptability, as the model must be retrained for each new city or policy setting. Furthermore, its applicability is restricted to cities where an agent-based simulation exists, making it less feasible for cities without pre-existing ABMs or with limited

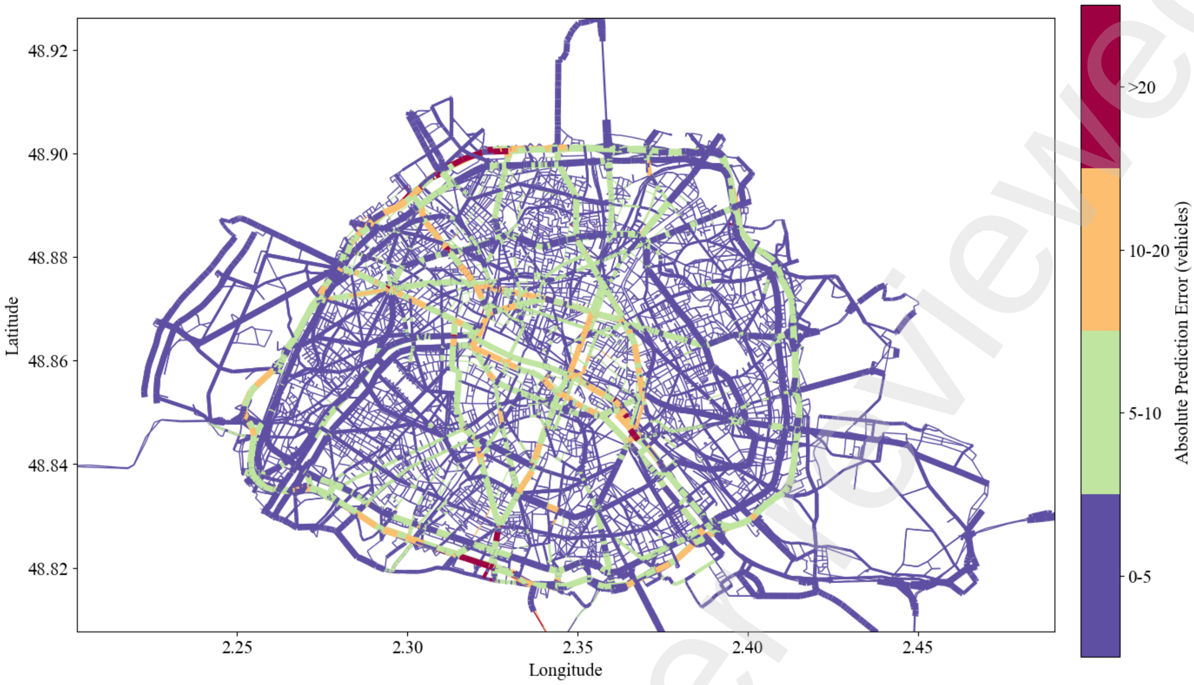


Figure 11 – *Absolute difference (MAE) between predicted and actual over all roads, over all observations in the test set*

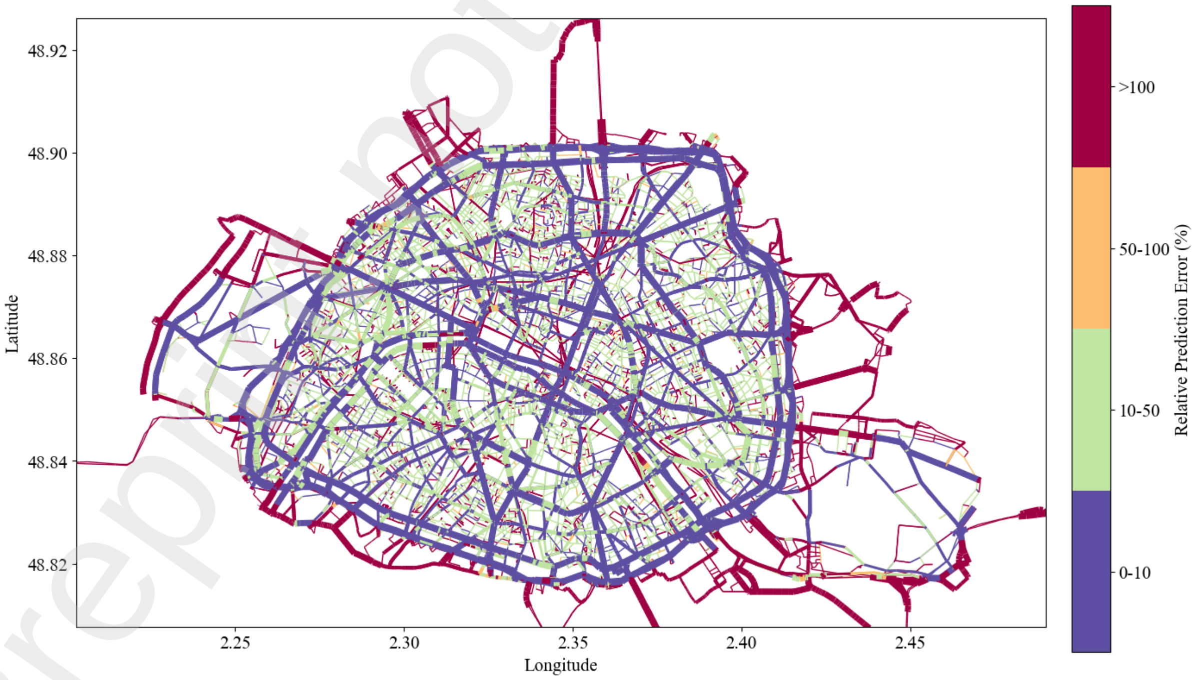


Figure 12 – *Relative difference (MAE) between predicted and actual over all roads, over all observations in the test set*

computational resources. Future work should explore inductive graph learning and domain adaptation techniques to enhance generalizability across different urban settings without requiring full retraining.

- *Neglect of Physical Traffic Principles*
  The surrogate model relies on learned representations rather than explicitly incorporating physical traffic flow constraints. While this allows flexibility and efficiency, it may overlook fundamental principles such as flow conservation, congestion spillover, capacity saturation, and queue propagation. Future research should explore hybrid approaches that integrate physics-based constraints with data-driven learning to enhance robustness.

- *Impact of Population Scaling*
  To balance computational efficiency and practical applicability, we trained the model using simulations scaled to 1% of households. While this approach enables feasible training, it introduces limitations. Smaller samples may distort travel time distributions and underrepresent certain travel behaviors, potentially affecting prediction accuracy. Studies suggest that a 5% or 10% scaling factor would improve fidelity, though at a higher computational cost. Future work should explore strategies to mitigate these effects, such as bias correction techniques or hybrid approaches that combine multiple scaling factors.

- *Interpretability and Transparency*
  Interpreting ML-based surrogate models remains a challenge, particularly in policy-sensitive applications. While GNNs effectively capture complex spatial dependencies, their decision-making processes lack transparency, making it difficult for policymakers to validate predictions. Enhancing model explainability through interpretable GNN architectures or post-hoc analysis techniques could improve trust and adoption.

Addressing these challenges will improve the adaptability and reliability of GNN-based surrogate models, ensuring their broader applicability in transport policy evaluation.

Despite its current limitations, the surrogate model's rapid evaluation capabilities could enable a range of applications in the long run. First, it could facilitate simulation-based optimization, allowing policymakers to efficiently explore large solution spaces and identify optimal policy setups. Second, its speed makes real-time control applications viable, enabling swift scenario evaluations in time-sensitive decision-making contexts. These capabilities can then help planners to systematically refine policy options to identify the most effective interventions and also provide a computationally efficient framework for objective evaluation in optimization-based approaches.

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10. Author contributions

The authors confirm their contributions to the paper as follows: E. Natterer, A. Tejada Lapuerta, R. Engelhardt, S. Hörl, and K. Bogenberger conceived and designed the study; Conzeptualization: E. Natterer, A. Tejada Lapuerta, R. Engelhardt; Methodology and Formal analysis: E. Natterer; Data curation, Software, Visualization: E. Natterer, S. R. Rao; Funding acquisition: K. Bogenberger. All authors reviewed the results and approved the final version of the manuscript.

11. Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used *Chat-GPT 4.0* in order to help with the writing process. Additionally, they used *perplexity.ai* and *answerthis.io* during the literature review. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A. Bias-Variance Decomposition

Appendix A.1. Definitions

Simulation Output

The change in traffic volume on each edge e in the network is stochastic due to randomness in the simulation. We model the true simulated change in traffic volume as:

$$y_e = \Delta b_e + \epsilon_e, \tag{A.1}$$

where:

- $\Delta b_e = b_e^{\text{policy}} - b_e^{\text{base}}$ represents the *deterministic change* in traffic volume caused by the policy. This is a fixed value representing the average impact of the policy on edge e.
- ϵ_e is the *random simulation noise*, representing the variability in the simulation output. This variability can be assumed to be the same for the base case and when introduced a policy, because the simulation remains stochastic under the policy.

Machine Learning Model Prediction

The machine learning model attempts to predict y_e, producing:

$$\hat{y}_e = y_e + \hat{\epsilon}_e. \tag{A.2}$$

We assume the prediction error is structured as:

$$\hat{\epsilon}_e = \epsilon_e + \epsilon'_e, \tag{A.3}$$

where:

- ϵ_e is the original simulation noise.
- ϵ'_e is the *additional error* introduced by the model. Note that this error is *deterministic*, as any output of a machine learning model.

Mean Squared Error (MSE)

The Mean Squared Error (MSE) is defined as:

$$\text{MSE}(y, \hat{y}) = \frac{1}{|E|} \sum_{e \in E} (y_e - \hat{y}_e)^2. \tag{A.4}$$

Substituting $\hat{y}_e = y_e + \hat{\epsilon}_e$, and $\hat{\epsilon}_e = \epsilon_e + \epsilon'_e$, the squared error simplifies to $(\epsilon'_e - \epsilon_e)^2$ and thus

$$\text{MSE}(y, \hat{y}) = \frac{1}{|E|} \sum_{e \in E} (\epsilon'_e - \epsilon_e)^2. \tag{A.5}$$

Appendix A.2. Introducing the Random Variable

When evaluating our machine learning surrogate, our goal is to predict, for each edge $e \in E$, the simulated change in traffic volume, y_e, using the model prediction $\hat{y}_e$. We measure this using the squared error between the simulated values y and the predicted values $\hat{y}$ for all edges $e \in E$. Because y_e includes a stochastic component ϵ_e, it varies between simulation repetitions. Hence, also the squared error varies. To summarize the model's performance in a consistent way, we model y_e as a random variable.

We define Y_e as the *random variable* representing the simulated change in traffic volume for edge $e \in E$, and $\hat{Y}_e$ as the *random variable* representing the predicted change in traffic volume for edge $e \in E$. Then, we can model $\mathbf{Y}$ as the random vector $\mathbf{Y} = (Y_{e_1}, Y_{e_2}, ...Y_{e_{|E|}})$, and similarly $\hat{\mathbf{Y}} = (\hat{Y}_{e_1}, \hat{Y}_{e_2}, ...\hat{Y}_{e_{|E|}})$.

Since we're now dealing with random variables, we can take the expected value of them. The expected value represents the average performance of the model across many potential outcomes. This ensures that our evaluation is fair and robust, as it accounts for the full range of variability in the simulation. By using the expected value, we evaluate how well the model performs on average over all possible realizations of the simulation. Thus, overall, we're interested in a lower bound of the following term:

$$\mathbb{E}[\text{MSE}(Y, \hat{Y})] = \frac{1}{|E|} \sum_{e \in E} \mathbb{E}[\text{MSE}(Y_e, \hat{Y}_e)] \tag{A.6}$$

where the equality follows straight away by the linearity of the expected value.

Therefore, let's focus on finding a lower bound for $\mathbb{E}[\text{MSE}(Y_e, \hat{Y}_e)]$. Note that Y_e, just like y_e, consists of a deterministic and a random part:

$$Y_e = \Delta b_e + \epsilon_e. \tag{A.7}$$

The deterministic change Δb_e is constant, and ϵ_e is the stochastic noise, which captures the random variability around the deterministic change. WLOG, we assume ϵ_e follows a normal distribution:

$$\epsilon_e \sim \mathcal{N}(0, \sigma_e^2). \tag{A.8}$$

Similarly, the machine learning model's prediction is:

$$\hat{Y}_e = Y_e + \hat{\epsilon}_e \tag{A.9}$$

where again $\hat{\epsilon}_e = \epsilon_e + \epsilon'$. ϵ' is a constant and thus has variance 0.

Thus,

$$\begin{aligned}
\mathbb{E}[\text{MSE}(Y_e, \hat{Y}_e)] &= \mathbb{E}[(Y_e - \hat{Y}_e)^2] &\text{(A.10)}\\
&= \mathbb{E}[(\epsilon_e + \epsilon_e')^2] &\text{(A.11)}\\
&= \mathbb{E}[\epsilon_e^2] + \mathbb{E}[(\epsilon_e')^2] + 2\mathbb{E}[\epsilon_e \epsilon'] &\text{(A.12)}\\
&= \mathbb{E}[\epsilon_e^2] + {\epsilon_e'}^2 + 2\epsilon' \mathbb{E}[\epsilon_e] &\text{(A.13)}\\
&= \sigma_e^2 + {\epsilon_e'}^2 &\text{(A.14)}
\end{aligned}$$

where Equation A.13 follows from Equation A.8 and that ϵ' is a constant.

Appendix A.3. Lower bound for the Mean Squared Error (MSE) in terms of the variance

From the computations above and plugging it into Equation A.6 we arrive at:

$$\mathbb{E}[\text{MSE}(Y, \hat{Y})] = \frac{1}{|E|} \sum_{e \in E} \sigma_e^2 + {\epsilon_e'}^2. \tag{A.15}$$

This represents the *best achievable error*. Even with perfect predictions of the deterministic component, the random variability in the simulation sets a limit on how low the error can go. That means, Equation A.15 holds even if the model perfectly predicts the simulation output $y_e = \hat{y}_e$ for all edges $e \in E$. Therefore, in general,

$$\mathbb{E}[\text{MSE}(Y, \hat{Y})] \geq \frac{1}{|E|} \sum_{e \in E} \sigma_e^2 + {\epsilon_e'}^2. \tag{A.16}$$

Appendix B. Analysis of Base Case Volume Patterns and Sampling Bias

Our analysis reveals a systematic pattern: roads that receive capacity reductions show higher base case volumes and variances compared to roads that never receive reductions. This pattern appears consistently across all road types (primary, secondary, and tertiary) and emerges despite our random district selection process. Here, we explain the underlying mechanisms that create this sampling bias.

Appendix B.1. The Pattern We Observe

For all road types, we consistently observe higher base case variance ($\sigma_{t,b}^2$) in roads with capacity reduction:

- Primary roads: $\sigma_{t,b}^2 = 134.29$ vs 112.78
- Secondary roads: $\sigma_{t,b}^2 = 63.16$ vs 49.35
- Tertiary roads: $\sigma_{t,b}^2 = 41.12$ vs 35.38

Appendix B.2. District Selection Process

Our methodology involves:

- We had 8,308 different scenarios available
- In each scenario, we randomly select about four districts
- All roads in a selected district with road type “primary”, “secondary” and “tertiary” receive capacity reduction
- A road is labeled as “with capacity reduction” if its district is selected in any scenario
- A road is labeled as “without capacity reduction” if its district is never selected

Appendix B.3. Analysis of District Characteristics

Our analysis of district-level data reveals several key patterns:

Appendix B.3.1. Volume Distribution Within Districts

- Districts show highly skewed volume distributions (mean skewness = 3.64)
- On average, only 27% of roads in a district have volumes above the district mean
- Volume distributions vary significantly between districts

Appendix B.3.2. Selection Frequency Patterns

When we rank districts by how often they are selected, we find clear patterns:

- Districts selected most frequently (top 25% by selection frequency):
  - Lower mean volumes (47.44)
  - Lower standard deviation (70.63)
  - More balanced distribution (skewness = 2.08)
  - Higher proportion of high-volume roads (30%)
- Districts selected least frequently (bottom 25% by selection frequency):
  - Higher mean volumes (60.37)
  - Higher standard deviation (158.44)
  - Highly skewed distribution (skewness = 5.76)
  - Lower proportion of high-volume roads (21%)

Appendix B.4. Key Statistical Relationships

Our analysis reveals two important correlations:

- Strong negative correlation (-0.762) between district volume skewness and selection frequency
- Positive correlation (0.605) between proportion of high-volume roads and selection frequency

Appendix B.5. How the Sampling Bias Emerges

The sampling bias develops through the following sequence:

1. **Initial District Selection:**
   - Districts with more balanced volume distributions tend to be selected more frequently
   - These districts have a higher proportion of high-volume roads
   - When selected, all roads in the district receive capacity reduction
2. **Cumulative Effect Over Many Scenarios:**
   - Roads from more balanced districts are more likely to receive capacity reduction
   - These districts contribute more high-volume roads to the ”with capacity reduction” group
   - This systematically increases both the average volume and variance in the capacity reduction group

Appendix B.6. Conclusion

The higher base case volumes and variances we observe in roads with capacity reduction are not a direct result of the capacity reductions themselves. Instead, they emerge from how our random selection process interacts with the underlying patterns of traffic volumes in different districts. The key insight is that districts with more balanced volume distributions are selected more frequently, and these districts tend to have a higher proportion of high-volume roads. This leads to their roads being overrepresented in the "with capacity reduction" group, creating the observed pattern in base case volumes and variances.